Journal of Financial Crises

Volume 4 | Issue 2

2022

Dividends and Bank Capital in the Global Financial Crisis of 2007–2009

Viral V. Acharya
New York University Stern School of Business

Irvind Gujral
Egon Zehnder

Nirupama Kulkarni
Centre for Advanced Financial Research and Learning (CAFRAL)

Hyun Song Shin
Bank for International Settlements

Recommended Citation
Acharya, Viral V.; Gujral, Irvind; Kulkarni, Nirupama; and Shin, Hyun Song (2022) "Dividends and Bank Capital in the Global Financial Crisis of 2007–2009," *Journal of Financial Crises*: Vol. 4 : Iss. 2, 1-39.
DOI: https://doi.org/10.17132/2693-3179.1290
Available at: https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss2/1

Dividends and Bank Capital in the Global Financial Crisis of 2007–2009

Viral V. Acharya,[*] Irvind Gujral,[†] Nirupama Kulkarni,[‡] Hyun Song Shin,[§]

Abstract

The headline numbers appear to show that even as banks and financial intermediaries suffered large credit losses in the Global Financial Crisis of 2007–2009, they raised substantial amounts of new capital, both from private investors and from government-funded capital injections. However, on closer inspection, the composition of bank capital shifted radically from one based on common equity to that based on debt-like hybrid claims such as preferred equity and subordinated debt. The erosion of common equity was exacerbated by large-scale payments of dividends, in spite of widely anticipated credit losses. Dividend payments represent a transfer from creditors (and potentially taxpayers) to equity holders in violation of the priority of debt over equity. The dwindling pool of common equity in the banking system may have been one reason for the continued reluctance by banks to lend over this period. We draw conclusions on how capital regulation may be reformed in light of our findings.

JEL classification: G01, G21, G24, G28

Keywords: capital requirements, common equity, dividends, financial crisis, irrational markets, share buy backs

* C.V. Starr Professor of Economics, New York University Stern School of Business. vacharya@stern.nyu.edu.

† Principal, Financial Services Practice, Egon Zehnder. irvind.gujral@gmail.com.

‡ Research Director, Centre for Advanced Financial Research and Learning (CAFRAL). nirupama.kulkarni@gmail.com.

§ Economic Adviser and Head of Research, Monetary and Economic Department, Bank for International Settlements. hyunsong.shin@bis.org.

I. Introduction

Financial intermediaries were at the center of the global financial crisis that began in August 2007. They bore the lion's share of the credit losses from securitized subprime mortgages, even though securitization was intended to parcel out and disperse credit risk to investors who were better able to absorb losses.[1] The capacity to lend suffered as intermediaries attempted to curtail their exposure to a level that could be more comfortably supported by their capital.[2] As the credit crisis deepened, banking assets across the board suffered. The pain was especially serious for subprime mortgages, commercial real estate, and household debt, ranging from credit card loans to auto loans.

The accumulated losses in the Global Financial Crisis (GFC) were very large, but so were the headline figures for the amount of new capital raised by financial institutions. Figures 1, 2, and 3 illustrate this "catching up" of capital with losses incurred.

Figure 1: Credit Losses and Write-downs Incurred (All Financial Firms Including Banks, Brokers, Insurers, and GSEs*), 2007–2009

	3Q07†	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09	Total
Worldwide	58.7	216.7	220.9	174.6	263.6	385.1	140.7	147.3	35.2	80.5	1,723.3
Americas	42.9	128.3	135.1	112.5	205.6	243.3	101.0	99.5	28.9	48.2	1,145.3
Europe	14.5	76.9	74.3	58.3	52.2	137.5	36.1	47.7	6.7	32.8	537.0
Asia	1.3	11.4	11.4	3.7	5.7	4.4	3.6	0.1	−0.4	−0.6	40.6

*GSEs are government-sponsored enterprises, hybrid public-private entities such as the Federal National Mortgage Association (Fannie Mae) in the US.

†All figures are in billions of USD.

Source: Bloomberg WDCI.

The cumulative acknowledged credit losses for financial institutions worldwide since the beginning of the GFC stood at $1.72 trillion. Set against this, the headline figure for new capital raised was $1.45 trillion. On the surface, the new capital raised was substantial, almost matching the losses. We see from Figures 1 and 2 that there were some regional variations, with new capital raised in the Americas being smaller relative to losses when compared to Europe. Although a substantial amount of new capital raised worldwide during the GFC was in the final quarter of 2008 as a part of government-funded recapitalization of the banking sector, the raw numbers are impressive.

[1] In some cases, this appears to have been by design, for example, in structured investment vehicles (SIVs) and asset-backed commercial paper (ABCP) conduits, where banks sold guarantees to securitization vehicles to game capital requirements. See Acharya, Schnabl, and Suarez (2009) for detailed evidence of such "securitization without risk transfer." In other cases, it appears to have been a highly levered bet on the economy, for example, as manifested in the holdings of AAA-rated mortgage-backed securities; banks held up to 39% of all such securities.

[2] Ivashina and Scharfstein (2010) document that during the crisis, especially in the aftermath of Lehman's collapse, banks made very few new loans and primarily honored drawdown on prearranged lines of credit.

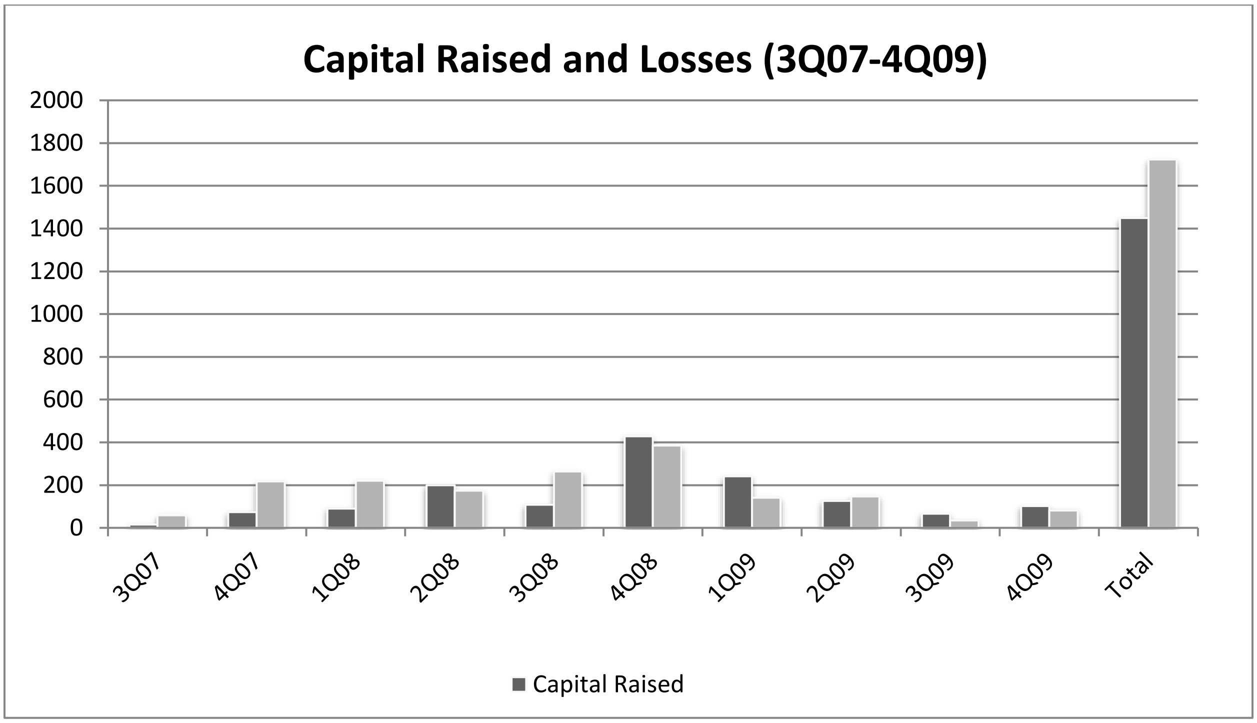
Figure 2: Capital Raised (WDCI—All Financial Firms Including Banks, Brokers, Insurers, and GSEs*), 2007–2009

($ bn)	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09	Total
Worldwide	14.8	74.1	89.7	199.4	107.7	427.9	241.1	125.8	66.8	101.6	1,448.9
Americas	3.1	47.1	63.0	103.8	44.1	266.6	122.0	91.8	16.1	21.1	778.7
Europe	11.7	26.9	23.0	82.0	54.9	132.7	99.6	13.7	30.7	75.1	550.3
Asia	0.0	0.0	3.7	13.5	8.7	28.6	19.5	20.3	20.0	5.4	119.7

* GSEs are government-sponsored enterprises, hybrid public-private entities such as the Federal National Mortgage Association (Fannie Mae) in the US.

Source: Bloomberg WDCI.

Figure 3: Capital Raised vs. Losses Incurred by Worldwide Financial Institutions



Source: Bloomberg WDCI.

However, a closer look at the numbers reveals a much less sanguine picture of the state of the banking sector. We focus our detailed descriptive analysis on the 25 largest financial firms (investment banks, commercial banks, and government-sponsored enterprises (GSEs)[3]) in the US, UK, and Europe during the GFC. Our analysis yields three observations, of these banks in particular, that are worthy of closer scrutiny.

[3] GSEs are government-sponsored enterprises, hybrid public-private entities such as the Federal National Mortgage Association (Fannie Mae) in the US.

First, even though losses on asset portfolios depleted the banking system's common equity, banks continued to pay dividends throughout the crisis. We show that the outflow of common equity in the form of dividends was substantial in relation to total assets and total credit losses. This outflow deprived the banking system of common equity capital precisely when it was most needed. This erosion of common equity through dividends points to a breakdown of the priority of debt over equity. Banks that ultimately received public funding support and were in serious risk of failure continued to pay out dividends right from the period leading up to the crisis until the period after Lehman Brothers' bankruptcy. For a bank whose losses could be anticipated, it can be argued that dividends were paid to equity holders at the expense of the debt holders (including the taxpayers who fund bailouts). This represents a straight transfer in violation of the priority of debt over equity, which is sustained because of the slow-moving nature of book equity. In effect, the inertia in bank accounting makes even a distressed bank appear healthy in terms of its book capital ratios, enabling a transfer in violation of the priority of debt over equity.[4] Eric Rosengren focused on this observation in his speech to the Federal Reserve Bank of Boston in 2010, where he remarked:. . . if dividends had been halted at the SCAP [Supervisory Capital Assessment Program] banks once the LIBOR [London Interbank Offered Rate] rate rose, nearly $80 billion would have been retained as capital. This represents close to 50 percent of the CPP funds used to recapitalize these banks in the fall of 2008. Clearly a proactive approach to dividend retention could have substantially reduced the need for an emergency infusion of public funds.

Adrian, Boyarchenko, and Shin (2016) empirically analyze these questions during the Global Financial Crisis and find continued dividend payouts. These dividend payouts continued in spite of lagging bank profitability in the presence of compressed net interest margins and low rates. Essentially, intermediaries managed the size of their book equity actively through their dividend payout decisions.

Second, most of the new capital being raised during the period was in the form of debt or hybrid claims such as preferred equity. We argue that this change in the composition of bank capital away from common equity to preferred equity potentially increased banks' leverage. To understand the significance of common equity and its role in bank resolution, we distinguish between two different notions of bank capital. In the first notion, bank capital acts as a buffer against loss that protects depositors. In this view, hybrid claims such as preferred equity or subordinated debt are counted as bank capital, since both are claims that are junior to depositors. A second contrasting notion of bank capital, which we term "pure equity capital," views bank capital as the claim held by the owners of the bank who have control over the bank's operations. Hybrid claims such as preferred shares or subordinated debt, in this view, do not qualify as bank capital and are junior forms of debt. Thus, when leverage is measured as the ratio of total assets to common equity, the leverage of the banking sector in the US, UK, and Europe rose relentlessly during the crisis, as we show below. We argue that the continued reluctance of banks to lend may have been attributable (at least in part) to the high leverage of the banking sector when measured using the pure

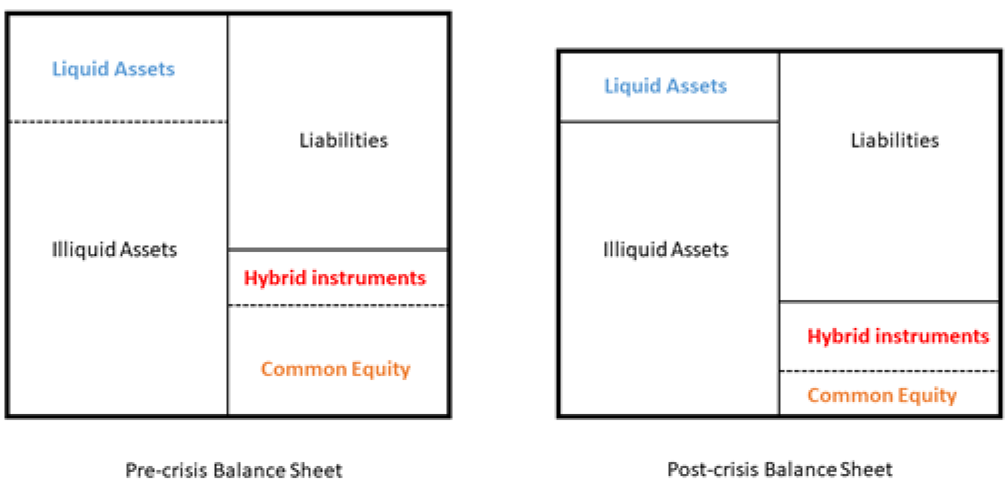
[4] The undesirable nature of dividend payments during crises has been commented on by Scharfstein and Stein 2008. See also Wessel 2008.

equity capital lens.

Last, but not the least, as common equity is paid out on the liabilities side of the balance sheet, the assets that get depleted on the asset side are the safe marketable assets—especially cash or government bond holdings. The illiquid, riskier assets end up left behind on the balance sheet. This implies a type of risk shifting, alternatively asset substitution. This risk shifting further favors the equity holders over the debt holders for the usual reason that equity holders' claims are convex claims over the asset payoffs, while debt holders have concave payoffs. Whereas, traditionally, risk shifting has been discussed mainly in the context of new investments (as in the seminal work of Jensen and Meckling 1976), we can see that risk shifting can also be accomplished through changes in the capital structure of the bank. Paying out dividends in the form of cash leaves behind riskier assets on a thinner equity cushion, which benefits the shareholders once again, at the expense of the debt holders.

As Figure 4 graphically shows, the change in composition of bank capital and the resulting deterioration of banks' balance sheets occur due to the shift to preferred equity and the erosion of common equity on the liability side of the balance sheet accompanied by the asset-side deterioration as safer liquid assets are depleted.

Figure 4: Liability-Side and Asset-Side Changes in Composition of Bank Balance Sheets during the 2007–2009 Global Financial Crisis



Note: The figures above graphically represent the change in composition of the asset side and the liability side of banks' balance sheets during the 2007–2009 crisis.

Source: Created by the authors.

On a related point, since many of the equity holders of banks are also employees, the diversion of funds from debt holders (including taxpayers) to equity holders is related to the thorny and politically charged issue of employee compensation in banks (Bae, Kang, and Wang 2011; Faleye, Mehrotra and Morck 2006). In this sense, our paper can be seen as a contribution pointing out how the determination of bank capital structure and dividend policy can be seen as a part of the larger debate on compensation issues. The standard view on corporate governance that emphasizes shareholder value maximization may have unintended and adverse consequences for the swift resolution of failing banks.

Our paper is primarily a descriptive study documenting the following in a comprehensive way: (1) the time profile of losses; and (2) the amount and type of new capital raised by banks, and especially since the beginning of the Global Financial Crisis. Although our study is by design a "fact-finding" study, we believe that it contributes on two fronts. First, the facts themselves are striking, and we have attempted to present the evidence in a unified way that conveys the big picture. Second, and more important, the facts uncovered provide insights both for the way that banks made decisions in the crisis and for future reform of the rules governing bank regulation.

In particular, we believe that the dwindling pool of common equity may have been an important reason for the continued reluctance of banks to extend credit in spite of the large-scale injection of bailout capital. Most of the public injections of bank capital in the US through the Troubled Assets Relief Program (TARP) took the form of preferred equity rather than common equity (even though, in some cases, preferred equity was ultimately converted to common equity). As a consequence, banks' leverage relative to common equity increased relentlessly. To the extent that the common equity cushion was subject to increasing compression, the stake of the controlling equity holders shrunk in accordance. This led banks to take an extremely conservative attitude toward taking up the slack in intermediation left by the collapse of the securitization market as they preferred to wait for the fortunes of their beleaguered assets and thinly capitalized balance sheets to resurrect rather than extinguish that option for lower-risk loans (see also Diamond and Rajan 2011 for a related theoretical point).

In a speech in 2009, Bill Dudley, then president of the Federal Reserve Bank of New York, noted that executives at banks and government-sponsored enterprises told regulators "repeatedly over the past 18 months" that "now is not a good time to raise capital." He went on to say:

> This desire to postpone capital raising stems in part from the fact that bank executives often do not want to dilute existing shareholders, which of course include themselves. [. . .] The self-interested thing to do is avoid the dilution and hope for a good state of the world.

The fear of dilution leads incumbent shareholders to underinvest in raising new common equity capital, an agency problem that is a variant of the Myers (1977) debt overhang problem[5] (again, not in the context of new investments).[6] This juxtaposition of agency

[5] Firms cut dividends to avoid the debt overhang problem (Myers 1977). A debt overhang emerges if a company is unable to make new investments on account of failure to raise new debt, as lenders are unwilling to lend to highly levered firms. Firms anticipating such opportunities maintain an equity cushion and thus cut dividends in times of distress.

[6] Some others, see, for example, Tucker (2008), argue that the reluctance may be due to banks wanting to avoid sending an adverse signal to markets and suffering dilution due to lemon's premium (as in the Myers and Majluf

problems at failing banks—underinvestment in issuance of new capital and erosion of existing capital through dividend distributions—poses some of the most difficult questions for bank resolution policy.

This divergence in the interests of the incumbent controlling shareholders from the broader public interest also raises questions on what the proper notion of regulatory capital should be. Under the current system of bank regulation, capital is regarded as a buffer against loss for senior creditors, and especially retail depositors. Hence, under the current system, regulatory capital includes subordinated debt and preferred equity. In future, regulators may have no choice but to employ intervention thresholds that are tied to the market value of equity—since that is what affects decisions of bank management—and market-imposed leverage constraints such as the extent of repurchase agreement (repo) "haircuts" faced by a financial institution in the market for borrowing.

Before we discuss these policy implications, we provide descriptive evidence on capital raised by 25 large banks in the US, the U. K., and Europe, and the Federal National Mortgage Association (Fannie Mae) and the Federal Home Loan Mortgage Corporation (Freddie Mac), two of the government-sponsored enterprises in the United States. We focus especially on the type of capital issued, and on the dividend policies and capital structure of these banks, in the period 2000–2009.[7]

The rest of the paper is organized as follows. In Section II, we provide evidence on the bank capital raised and dividends paid during 2007–2009 and contrast this with pre-crisis trends in 2000–2006. In Section III, we explore the difference between Basel capital and pure equity capital and demonstrate why the newly raised capital was mostly in the form of debt-like hybrid claims. In Section IV, we explore the differential payouts of commercial versus investment banks. In Section V, we look at the implications of the analysis for financial reform. Section VI concludes.

II. Evidence on Bank Capital and Dividends

Adrian and Shin (2007) document, and we find, that the significant balance sheet growth of banks between 2000 and 2006 was funded by a combination of (primarily) debt and (to a lesser extent) preferred equity. Figure 5 shows the total capital raised by each financial firm[8] in our sample, during the period of 2000 to 2006, by the type of instrument—common equity, preferred equity, or debt. For the period preceding the crisis, a total of $1.38 trillion

---

[1984] model of costly equity issuance). Tucker also raises the possibility that bankers may believe that many of their assets are in fact "under-valued" and thus, may be avoiding accumulating excess capital. However, he hastens to add that this belief is at odds with the almost perfect dearth of buyers for these supposedly undervalued assets.

[7] Complete details are provided in the Appendixes. Appendixes A and B describe the variables we employ, their sources, and the frequency of their measurement.

[8] Our sample of the 25 largest financial firms includes US/UK/European banks and the GSEs Fannie Mae and Freddie Mac. For ease of exposition, we refer to them collectively as financial firms or as banks.

of capital was issued by the 25 large financial firms in our sample. Despite the large amount of capital raised, a staggering $1.41 trillion (102% of capital) in funding was also raised in the form of debt. Preferred equity accounted for $46.3 billion (3%). Capital outflow to common shareholders was at $75 billion (5%). Common equity is the equity issuance net of buybacks. Thus, overall, banks were in fact net buyers of common equity rather than issuers. During this period, the 25 large financial firms of the US, UK, and Europe, had negative common equity issuance—that is, they bought back more shares, $116 billion more, than they issued. This pattern is remarkable since this was a period during which bank balance sheets grew significantly. Figure 6 shows the total capital raised and loss incurred by each financial firm[9] in our sample, during the period Q3'07–Q4'09, by the type of instrument—common equity, preferred equity, or debt. Figure 7 shows the total quarterly capital raised by each financial firm[10] in our sample, during the period Q3'07–Q4'09.

Figure 5: Capital Raised (Lost), by Type of Instrument, for 25 Large Financial Firms, 2000–2006

Geography	Name	Type of instrument						Total capital raised
		Common		Preferred		Debt		
		($ bn)	% of total capital raised	($bn)	% of total capital raised	($ bn)	% of total capital raised	($ bn)
US	JPMorgan	−3.0	−6%	−2.2	−4%	54.1	110%	49.0
US	Wells Fargo	−9.0	−18%	2.9	6%	57.4	112%	51.3
US	Lehman Brothers	0.7	1%	0.5	1%	52.5	98%	53.7
US	Wachovia Corp.	−12.1	−57%	0.0	0%	33.2	157%	21.1
US	Citigroup	7.1	4%	-0.9	-1%	153.5	96%	159.7
US	Washington Mutual	−9.4	191%	2.9	-60%	1.6	−32%	−4.9
US	Merrill Lynch	−9.0	−8%	2.7	3%	112.6	106%	106.3
US	Morgan Stanley	−12.5	−13%	3.1	3%	105.8	110%	96.3
US	Bank of America	−34.2	−162%	2.6	12%	52.7	250%	21.1
US	Goldman Sachs	−17.7	−14%	3.1	2%	139.6	112%	124.9
US	Fannie Mae	−1.7	−1%	0.4	0%	220.6	101%	219.3
US	Freddie Mac	−2.0	−11%	1.5	8%	18.2	103%	17.6
US	Bear Stearns	−1.0	−4%	0.0	0%	27.4	104%	26.3
UK	Royal Bank of Scotland	10.8	23%	12.4	27%	22.8	50%	46.0
UK	HSBC	4.6	19%	7.1	29%	12.8	52%	24.5
UK	Barclays Plc	−1.9	−13%	-0.4	−3%	16.4	117%	14.1
UK	HBOS	2.1	8%	1.5	6%	21.9	86%	25.5
UK	Lloyds TSB	0.8	11%	0.0	0%	7.1	90%	7.9
Europe	IKB	7.1	23%	8.3	27%	15.5	50%	31.0

9 Our sample of the 25 largest financial firms includes US/UK/European banks and the GSEs Fannie Mae and Freddie Mac. For ease of exposition, we refer to them collectively as financial firms or as banks.

10 Our sample of the 25 largest financial firms includes US/UK/European banks and the GSEs Fannie Mae and Freddie Mac. For ease of exposition, we refer to them collectively as financial firms or as banks.

Europe	UBS	−2.7	−3%	0.0	0%	92.7	103%	90.0
Europe	Credit Suisse	5.7	8%	0.0	0%	61.3	92%	66.9
Europe	Deutsche Bank	2.0	3%	0.9	1%	72.0	96%	74.9
Europe	Fortis Bank	0.5	1%	0.0	0%	53.5	99%	53.9
	TOTAL	**−74.8**	**−5%**	**46.3**	**3%**	**1,405.2**	**102%**	**1,376.5**

Note: Data is not available for BNP Paribas and ABN AMRO.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, and Bloomberg.

Figure 6: Total Capital Raised or Lost, by Type of Instrument, and Losses Incurred for 25 Large Financial Firms, Q3 ’07–Q4 ’09

Geography	Name	Type of instrument						Total capital raised	Losses incurred*
		Common		Preferred		Debt			
		($ bn)	% of total capital raised	($ bn)	% of total capital raised	($ bn)	% of total capital raised	($ bn)	($ bn)
US	JPMorgan	44.0	36%	28.7	23%	51.1	41%	123.8	62.8
US	Wells Fargo	28.8	273%	22.7	215%	−40.9	−389%	10.5	43.0
US	Lehman Brothers	0.1	0%	5.9	10%	50.1	89%	56.0	16.2
US	Wachovia Corp.	3.1	5%	9.7	17%	44.8	78%	57.5	101.8
US	Citigroup	25.4	28%	69.6	76%	−3.2	−3%	91.9	123.9
US	Washington Mutual	−1.5	43%	9.4	−267%	−11.4	324%	−3.5	45.1
US	Merrill Lynch	9.4	18%	10.4	20%	32.3	62%	52.1	55.9
US	Morgan Stanley	1.9	4%	8.0	15%	44.0	81%	54.0	23.4
US	Bank of America	20.9	21%	66.3	66%	13.7	14%	100.9	89.2
US	Goldman Sachs	2.8	4%	3.8	6%	55.5	89%	62.1	9.2
US	Fannie Mae	16.1	3,318%	15.3	3,161%	−30.9	−6,379%	0.5	138.7
US	Freddie Mac	−1.0	3%	7.9	−26%	−36.7	123%	−29.8	115.1
US	Bear Stearns	−1.5	−13%	0.0	0%	13.3	113%	11.8	3.2
UK	Royal Bank of Scotland	0.2	−4%	−0.9	17%	−5.0	87%	−5.7	56.7
UK	HSBC	28.8	63%	0.0	0%	17.1	37%	45.9	55.8
UK	Barclays Plc	16.2	55%	0.0	0%	13.1	45%	29.4	39.7
UK	HBOS	4.8	43%	0.0	0%	6.5	57%	11.3	26.3
UK	Lloyds TSB	36.2	109%	0.0	0%	−3.0	−9%	33.2	3.2
Europe	IKB	0.2	−3%	−1.9	33%	−4.1	70%	−5.8	12.5
Europe	UBS	0.0	0%	0.0	0%	54.2	100%	54.2	57.0
Europe	Credit Suisse	2.4	9%	0.0	0%	25.2	91%	27.7	18.9
Europe	Deutsche Bank	0.6	14%	8.7	205%	−5.1	−119%	4.3	19.5
Europe	Fortis Bank	25.1	62%	0.0	0%	15.4	38%	40.5	8.0
	TOTAL	**262.9**	**32%**	**263.6**	**32%**	**296.1**	**36%**	**822.6**	**1,146.1**

* Losses incurred for the crisis period from Q3 2007 to Q4 2009, quarterly breakdown is shown in Table 5. Capital raised data is not available for BNP Paribas and ABN AMRO.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

Figure 7: Quarterly Capital Raised (in $ bn), by Large Financial Firms, Q1 ’07–Q4 ’09

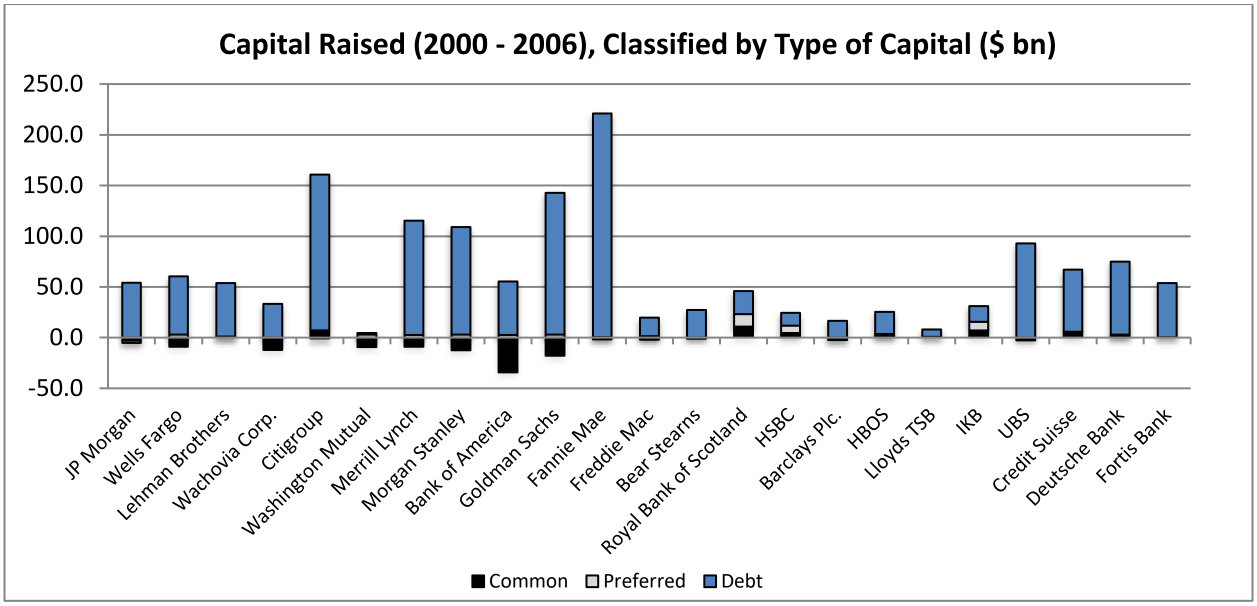
Geography	Name	1Q07	2Q07	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09	Total capital raised
US	JPMorgan	1.2	13.0	21.4	–0.2	2.3	18.5	4.0	62.3	–1.2	–14.8	–4.2	21.6	123.8
US	Wells Fargo	2.0	2.6	–0.1	1.1	0.5	2.1	4.8	33.6	–13.6	-8.4	-19.0	4.9	10.5
US	Lehman Brothers	12.8	10.7	16.4	0.2	8.1	7.8	—	—	—	—	—	—	56.0
US	Wachovia Corp.	3.6	–0.9	16.4	4.7	18.1	16.7	–1.1	—	—	—	—	—	57.5
US	Citigroup	9.8	16.6	13.5	13.2	11.8	12.7	–18.9	29.4	–8.7	2.2	13.1	–2.8	91.9
US	Washingt on Mutual	–1.5	1.6	–1.1	–0.8	–7.8	6.0	0.0	0.0	0.0	0.0	0.0	0.0	–3.5
US	Merrill Lynch	23.7	20.4	30.1	–1.8	–1.0	19.5	–18.6	–20.2	0.0	0.0	0.0	0.0	52.1
US	Morgan Stanley	14.5	9.8	3.5	10.8	5.2	8.8	–1.8	–2.9	5.0	–4.7	2.7	7.6	58.6
US	Bank of America	4.7	17.4	13.3	1.8	13.1	13.4	–9.1	36.1	19.5	6.3	0.1	–16.0	100.9
US	Goldman Sachs	11.3	10.9	19.2	8.0	11.7	8.2	–3.1	4.8	6.3	–5.3	–2.5	–7.3	62.1
US	Fannie Mae	5.0	10.0	–7.5	–38.3	–97.4	101.0	–8.4	–11.6	54.8	12.4	–0.8	–18.6	0.5
US	Freddie Mac	0.0	11.2	–13.9	–29.9	16.8	35.5	364.1	–443.0	78.2	–16.9	–12.6	–19.5	–29.8
US	Bear Stearns	4.1	2.8	3.2	0.8	0.8	—	—	—	—	—	—	—	11.7
UK	Royal Bank of Scotland	—	—	—	5.9	—	—	—	1.1	—	—	—	–12.7	–5.7
UK	HSBC	—	—	—	10.1	—	—	—	9.1	—	—	—	26.7	45.9
UK	Barclays Plc	—	—	—	9.2	—	—	—	20.3	—	—	—	–0.1	29.4
UK	HBOS	—	—	—	6.5	—	—	—	4.8	—	—	—	0.0	11.3
UK	Lloyds TSB	—	—	—	0.2	—	—	—	1.5	—	—	—	31.5	33.2
Europe	IKB	—	—	—	4.4	—	—	—	1.0	—	—	—	–11.2	–5.8
Europe	UBS	—	—	—	42.8	—	—	—	9.5	—	—	—	2.0	54.2
Europe	Credit Suisse	—	—	—	14.0	—	—	—	23.0	—	—	—	–9.3	27.7
Europe	Deutsche Bank	—	—	—	-0.8	—	—	—	4.6	—	—	—	0.4	4.3
Europe	Fortis Bank	—	—	—	46.7	—	—	—	-0.8	—	—	—	–5.4	40.5
	TOTAL	**91.3**	**126.1**	**114.6**	**108.6**	**–17.7**	**250.0**	**312.0**	**–237.4**	**140.4**	**–29.3**	**–23.1**	**–8.1**	**827.2**

Note: Capital raised data is not available for BNP Paribas and ABN AMRO.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

Figure 8 plots this division of capital issued by security type for individual banks for the period preceding the crisis (from 2000 to 2006). Our initial observations are that the firms that had negative debt to equity (representing capital outflow) were the ones that performed the worst during the crisis. Of note, JPMorgan, Wachovia, Wells Fargo, Bear Stearns, Merrill Lynch, Morgan Stanley, Bank of America, Goldman Sachs, UBS, Fannie Mae, and Freddie Mac had negative debt-to- equity ratios (indicating a capital outflow to common equity holders). HBOS, one of the beleaguered UK banks during the crisis, had a debt-to-common-equity ratio of 10.43. Even with the benefit of hindsight, the relationship between type of capital issued and the ex-post performance of banks is hard to ignore.

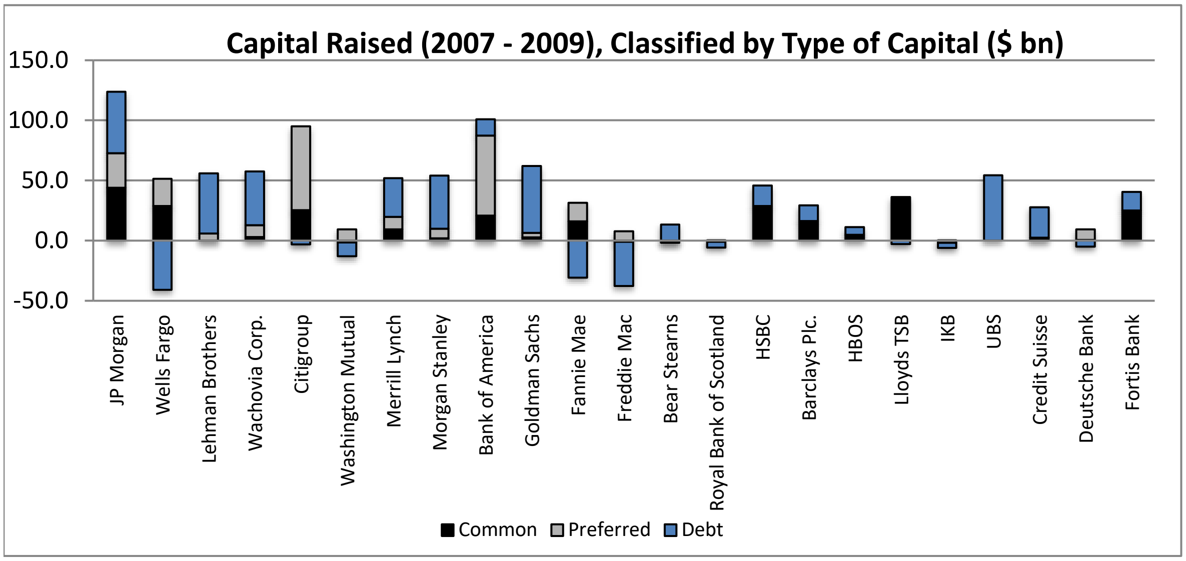
Figure 8: Capital Raised, Classified by Type of Instrument, for 25 Large Financial Firms, 2000–2006



Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

During the crisis period from 2007–2009 (see Figure 6 and 7), the large financial firms raised nearly $823 billion in capital. Figure 10 shows the comparison in the percentage of capital raised from debt, common equity, and preferred equity for the pre-crisis period (2000–2006) and crisis period (2007–2009). The proportion of debt fell to 36% of total capital and accounted for $296 billion of total capital. The common equity share stood at $263 billion and accounted for 32% of capital. In contrast to pre-crisis trends, where only 3% was issued in the form of preferred shares, more than 32% ($264 billion) of capital was issued in the form of preferred shares between 2007 and 2009.

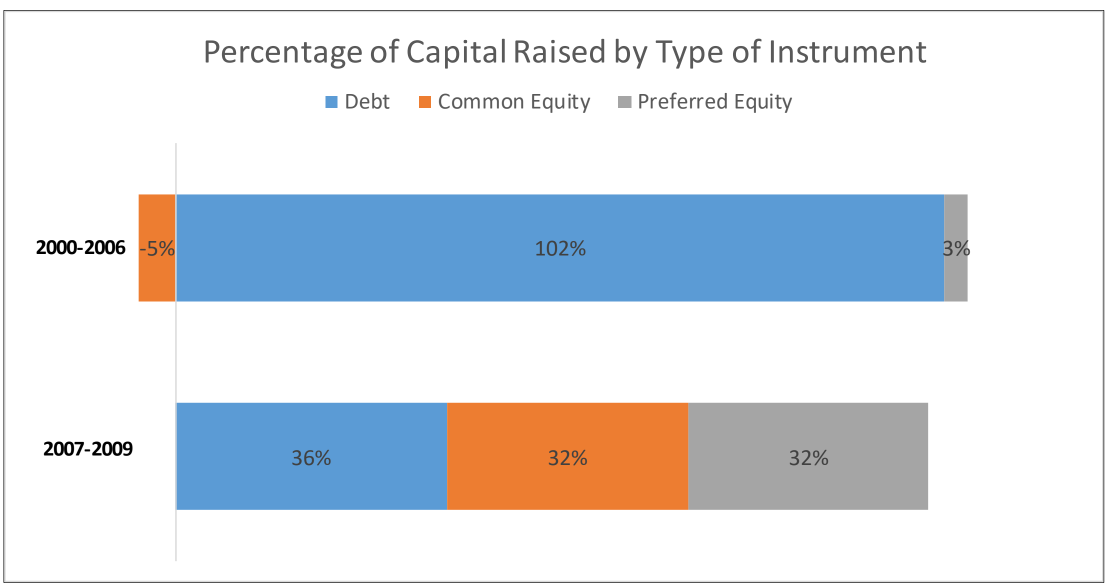
Figure 9: Capital Raised, Classified by Type of Instrument, for 25 Large Financial Firms, 2007–2009



Note: No data is available for BNP Paribas and ABN AMRO.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

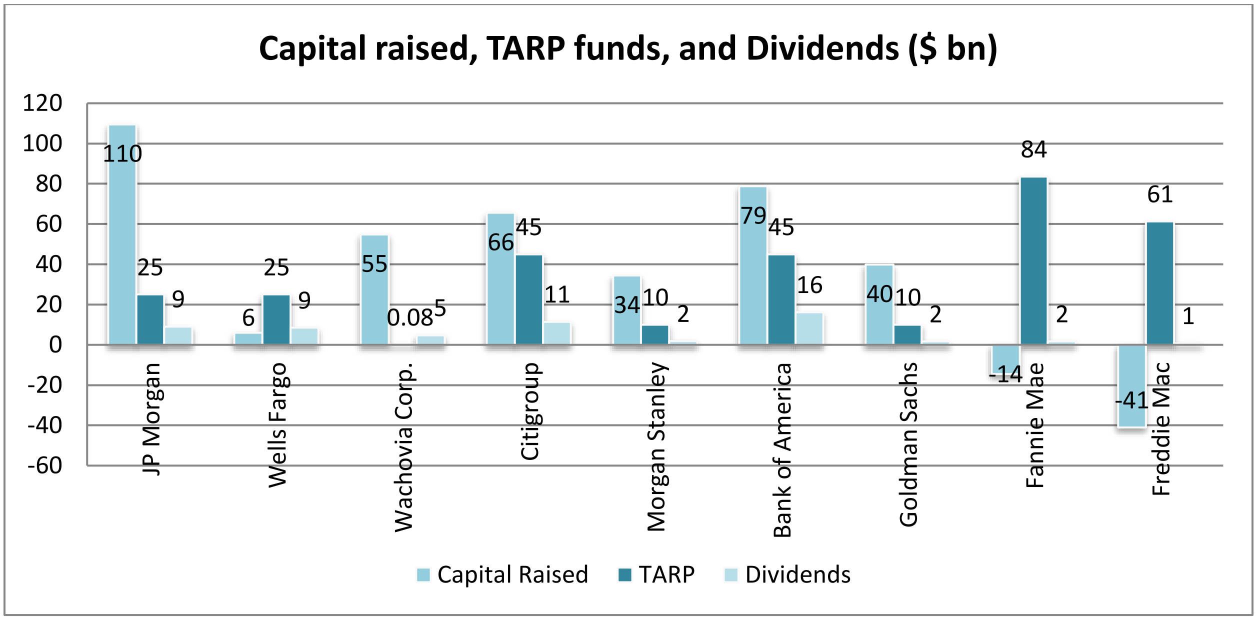
Figure 10: Capital Raised, Classified by Type of Instrument for 25 Large Financial Firms, for the Periods from 2000–2006 and 2007–2009



Sources: Bloomberg, US Securities and Exchange Commission, annual reports, Datastream.

Figure 11 shows the TARP funds received by banks and the total amount paid back from Q3 '07 to Q4 '09. The figure shows some striking results. All the banks (excluding the GSEs) that received TARP funding paid at least 34% of the amount received in TARP funds as dividends in this period. JPMorgan paid out $9 billion dollars, 36% of the TARP funds it eventually received from the government. Similarly, Bank of America and Citigroup, each of which received $45 billion in TARP funds, paid out $16 billion and $11 billion, respectively, between Q3 '07 and Q4 '09 (also refer to Figure 7 for data on quarterly capital raised by large financial firms).

Figure 11: Capital Raised, TARP Funds Received, and Dividends Paid for Large Financial Firms in the US, Q3 '07–Q4 '09



Sources: Annual statements of banks, US Securities and Exchange Commission filings, Bloomberg.

Figures 12, 13, 14, and 15 show that the evidence thus far masks one important fact—banks had been paying out significant dividends (measured as a percentage of total assets), not just during 2000–2006 but also during the crisis period of 2007–2009, particularly toward the beginning of the crisis. In effect, bank management did not drastically reduce their dividends in the first 12 months of the worst crisis to have hit them. From Figure 12, we see that US banks maintained quarterly dividends (measured as a percentage of total assets) between 0.09% and 0.06% from Q3 '07 to Q4 '08, before cutting dividends to 0.01% in 2009. In the pre-crisis period of 2000 to 2006, dividends averaged 0.08% total assets. Non-US banks cut dividends earlier, beginning in 2008 (see Figure 13). When we look at the combined trends in dividends for all 12 non-US banks in our sample, bank dividend payouts, measured as a percentage of combined total assets, were at 0.26% in 2002. In 2008, during the peak of the crisis, dividend levels fell to 0.17% and eventually to 0.05% in 2009.

Figure 12: Quarterly Dividends Paid by US Banks and GSEs, as Reported on the Banks' Balance Sheets

($bn)	2000–2006 per-quarter average	1Q07	2Q07	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09
Total dividends paid	6.38	10.45	10.64	11.21	11.19	9.71	9.20	8.45	6.35	1.97	0.72	0.78	0.78
Quarterly dividends as % of assets*	0.08%	0.09%	0.08%	0.09%	0.09%	0.07%	0.07%	0.07%	0.06%	0.02%	0.01%	0.01%	0.01%

* Calculated as total dividends paid by all banks as a percentage of the sum of assets of all banks.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

Figure 13: Semiannual Dividends Paid by Non-US Banks, as Reported on the Banks' Balance Sheets

($ bn)	2000–2006 per half-yearly average	1H07	2H07	1H08	2H08	1H09	2H09
Total dividends paid	5.98	12.57	30.11	6.99	11.77	2.80	8.71
Semiannual dividends as % of assets*	0.07%	0.07%	0.14%	0.03%	0.06%	0.01%	0.05%

*Calculated as total dividends paid by all banks as a percentage of the sum of assets of all banks.

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Compustat, and Bloomberg.

Figures 14 and 15 show the dividends paid by the 25 large financial institutions. We highlight the differences in dividend payouts by investment banks compared to commercial banks. The largest dividends were paid by Bank of America and Citigroup, with their dividends showing no slowdown through Q3 ’08. While dividend payments slowed for both banks from Q4 ’08 to Q4 ’09, Bank of America continued to pay dividends till the end of Q4 ’09. Citigroup, on the other hand, cut its dividends only in Q4 ’08 and paid zero dividends from Q2 ’09 to Q4 ’09.

Figure 14: Quarterly Dividends Paid by Each US Bank and GSE (Numbers from Balance Sheets)

($ mm)	JPMorgan	Wells Fargo	Lehman Brothers	Wachovia Corp.	Citigroup	WaMu	Merrill Lynch	Morgan Stanley	Bank of America	Goldman Sachs	Fannie Mae	Freddie Mac	Bear Stearns
2000–2006	25,603	19,438	1,053	14,879	42,237	9,876	4,308	5,107	38,756	2,632	2,741	2,848	2,943
1Q07	1,197	948	81	1,071	2,682	477	294	272	2,502	163	390	335	38
2Q07	1,328	937	81	1,066	2,671	484	292	269	2,494	161	490	326	38
3Q07	1,320	1,034	81	1,215	2,690	486	288	271	2,829	150	489	324	37
4Q07	1,320	1,036	81	1,265	2,690	482	361	270	2,830	165	487	167	36
1Q08	1,326	1,024	94	1,274	1,676	130	341	276	2,859	157	344	162	47
2Q08	1,362	1,026	95	808	1,753	10	344	280	2,858	156	343	162	—
3Q08	1,462	1,128	118	108	1,746	—	469	281	2,929	155	54	0	—
4Q08	1,483	1,134	—	107	875	—	699	273	1,610	174	0	0	—
1Q09	242	1,443	—	—	54	—	0	0	64	167	0	0	—
2Q09	163	214	—	—	0	—	0	80	86	180	0	0	—
3Q09	207	234	—	—	0	—	0	65	88	184	0	0	—
4Q09	208	238	—	—	0	—	0	65	88	186	0	0	—
2007–2009	11,618	10,396	—	—	—	—	—	—	—	—	2,597	1,457	196

Source: Bloomberg, SEC, annual reports, Datastream, Compustat

Figure 15: Semiannual Dividends Paid by Each Non-US Bank (Numbers from Balance Sheets)

($ mm)	Royal Bank of Scotland	HSBC	Barclays Plc	HBOS	Lloyds TSB	IKB	UBS	Credit Suisse	Deutsche Bank	Fortis Bank	BNP Paribas	ABN AMRO
2000–2006	20,169	44,839	16,540	11,451	21,799	512	2,625	5,605	8,596	10,397	12,936	12,058
1H07	1,911	3,982	1,440	1,226	1,245	—	0	0	0	1,269	0	1,497
2H07	4,215	6,511	3,094	2,443	2,833	—	0	2,328	3,091	1,435	4,159	0
1H08	0	2,113	1,445	—	1,280	—	0	0	0	2,154	0	0
2H08	4,002	5,600	301	—	0	—	0	109	420	0	1,342	0
1H09	0	2,800	0	—	0	—	0	0	0	0	0	0
2H09	0	3,101	288	—	0	—	0	2,190	650	0	2,479	0
2007–2009	10,129	24,107	6,569	3,669	5,358	—	0	4,627	4,161	4,859	7,980	1,497

Sources: Annual statements of banks, US Securities and Exchange Commission filings, Bloomberg, and Compustat.

Merrill Lynch almost doubled its dividends in Q4 ’08 (to $699 million) compared to the year earlier in Q4 ’07 ($361 million). Similarly, Lehman increased its dividends from $95 million in Q2 ’08 to $118 million in Q3 ’08, right before it went bankrupt. Bear Stearns also increased dividends from $36 million in Q4 ’07 to $47 million in Q1 ’08. Of note is Goldman Sachs, which continued to pay dividends until the end of 2009. It *increased* its dividends from $639 million in 2007 (annual) to $642 million in 2008 (annual) and to $717 million in 2009 (annual). On the other hand, while Morgan Stanley cut its dividends to zero in Q1 ’09, it resumed paying dividends in Q2 ’09. However, these Q2 ’09 dividends were only $80 million, 28% of dividends paid out in Q2 ’08.

In contrast to investment banks, commercial banks such as Wachovia and Washington Mutual cut their dividends drastically in the quarters leading up to their failure. Wachovia cut its dividends from $808 million in Q2 ’08 to $108 million in Q3 ’08. Similarly, Washington Mutual cut its dividends from $130 million in Q1 ’08 to $10 million in Q2 ’08. These numbers highlight the difference in dividend payouts between investment banks and commercial banks. Similarly, the GSEs, Fannie Mae and Freddie Mac, cut their dividends to zero in Q4 ’08 and Q3 ’08, respectively.

Figure 16 gives the quarterly losses incurred by the financial firms in our analysis. This table highlights the fact that these financial firms were struggling during this period and yet continued to pay dividends as described above. Particularly, Lehman, which increased dividends in Q3 ’08, posted losses of $5.3 billion in Q2 ’08 and $7.0 billion in Q3 ’08, before filing for bankruptcy. Bear Stearns, which increased dividends in Q1 ’08, posted losses of $1.9 billion in Q4 ’07 and $600 million in Q1 ’08 before being bought out by JPMorgan. Fannie Mae, which decreased its dividend in 4Q ’07, posted losses of $138.7 billion for the period from Q3 ’07 to Q4 ’09. Freddie Mac decreased its quarterly dividends steeply in 3Q ’07and posted losses of $115.1 billion for the period from Q3 ’07 to Q4 ’09. Wachovia, which cut dividends in Q3 ’08, reported $29.4 billion in losses for that same quarter, a jump of 124% from a loss of $13.1 billion in the previous quarter. Similarly, Washington Mutual, which cut dividends in Q2 ’08, reported losses of $30.9 billion in Q3 ’08, representing a 462% increase from $5.5 billion in losses in the previous quarter.

Figure 16: Quarterly Losses (in $ bn) Incurred by Large Financial Firms, Q3 2007–Q4 2009

	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09	Total
JPMorgan	2.5	2.8	5.9	4.0	8.1	9.8	7.7	8.0	7.8	6.2	62.8
Wells Fargo	0.0	2.6	2.2	4.5	5.5	8.6	4.3	4.7	5.6	5.0	43.0
Lehman	0.7	0.8	2.4	5.3	7.0	—	—	—	—	—	16.2
Wachovia	1.7	3.3	4.5	13.1	29.4	49.8	0.0	0.0	0.0	0.0	101.8
Citigroup	5.6	18.2	19.6	12.2	12.8	19.7	13.8	10.3	5.6	6.1	123.9
WaMu	0.9	3.9	3.9	5.5	30.9	—	—	—	—	—	45.1
Merrill	9.4	18.0	7.6	8.9	12.0	0.0	—	—	—	—	55.9
Morgan Stanley	0.9	9.4	2.3	1.8	1.3	5.8	1.2	0.2	0.4	0.1	23.4
Bank of America	2.1	7.6	6.3	5.5	6.7	14.5	13.8	13.3	10.1	9.3	89.2
Goldman Sachs	1.5	-0.5	2.0	0.8	1.1	2.3	0.8	1.2	0.0	0.0	9.2
Fannie Mae	3.7	5.2	9.5	15.4	20.0	17.0	19.3	24.0	13.5	11.1	138.7
Freddie Mac	3.1	6.1	15.2	13.0	19.0	22.2	13.5	24.5	-6.1	4.6	115.1
Bear Stearns	0.7	1.9	0.6	—	—	—	—	—	—	—	3.2
RBS	0.0	2.8	0.0	9.9	0.3	17.2	0.0	15.8	1.0	9.7	56.7
HSBC	0.9	7.6	2.6	10.8	4.8	15.6	0.0	7.9	0.0	5.6	55.8
Barclays	0.0	3.5	1.5	4.4	0.0	11.1	3.4	8.2	0.0	7.6	39.7
HBOS	0.0	1.1	4.2	0.2	3.7	17.1	—	—	—	—	26.3
Lloyds TSB	0.0	0.4	—	—	—	—	—	—	—	—	—
IKB	0.0	0.0	12.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	12.5
UBS	4.7	14.6	19.5	6.0	4.7	4.0	3.5	0.2	-0.1	-0.1	57.0
Credit Suisse	1.9	4.0	5.3	0.0	2.9	3.3	1.4	0.0	0.1	0.0	18.9
Deutsche	2.6	0.1	3.3	2.8	2.6	3.8	1.7	1.2	0.6	0.8	19.5
Fortis	0.0	4.5	2.7	0.8	0.0	0.0	0.0	0.0	0.0	0.0	8.0
BNP	0.5	1.3	1.0	1.0	2.6	3.7	2.4	2.1	2.5	2.0	19.1
ABN AMRO	0.0	1.9	—	—	—	—	—	—	—	—	1.9

Sources: Bloomberg WDCI (numbers as of December 31, 2009).

In summary, the analysis in this section highlights three important points. First, while it appears that banks raised capital as the crisis progressed, a large part of the newly raised capital came from debt-like hybrid claims such as preferred equity and subordinated debt. Second, the erosion of common equity was exacerbated by large-scale payments of dividends (particularly toward the first half of the crisis). Third, these effects were particularly striking for the investment banks compared to the commercial banks.

III. Two Notions of Capital

To understand the significance of common equity and its role in bank resolution, it is important to distinguish between two different notions of bank capital.[11] There is, first, the

[11] Marc R. Saidenberg and Til Schuermann (2003) explains the New Basel Capital Accord for bank capital regulation.

notion of bank capital (implicit in the Basel approach) as a buffer against loss that protects depositors. Under this notion of bank capital, which we will call "Basel capital," hybrid claims such as preferred equity or subordinated debt are counted as bank capital, since both are claims that are junior to depositors. Indeed, under the Basel capital accords, subordinated debt counts as Tier 2 capital. However, there is a second, contrasting notion of bank capital as the claim held by the owners of the bank who have control over the bank's operations. Hybrid claims such as preferred shares or subordinated debt do not qualify as bank capital under this second notion of bank capital, as they can be seen as junior forms of debt. We could dub this second notion of capital "pure equity capital." This notion of capital can be thought of as the equity demanded by creditors as a safeguard against losses on their stakes. It is analogous to the margin requirement set by creditors on leveraged traders and is exemplified by the "haircut" demanded by creditors in a repurchase agreement. In contrast to the Basel capital requirement (which is a regulatory capital requirement), we could characterize the pure equity capital requirement in the margin or haircut set by a creditor as the "market-determined" capital requirement. Just as with repo haircuts and margin requirements, the "pure equity" capital requirement fluctuates over time with shifts in market conditions and the balance sheet capacity of leveraged traders. The key difference between Basel capital as a buffer to protect depositors and pure equity capital as the market-determined haircut lies in the behavior of those owners who have control over the bank. When the bank has too little pure equity capital, the owners' incentives reflect their highly leveraged balance sheet. When faced with a dwindling stake in a leveraged entity, controlling owners have little to lose, and everything to gain by engaging in risk-shifting bets on the bank. The increased haircut imposed by the capital market during distress episodes could be seen as the increased margin demanded by creditors in the capital market to changed incentives, or the reduction in funding capacity of an asset in anticipation of the attendant risk-shifting problem.[12]

The market-determined capital requirement reflected in the repo haircut is a constraint imposed by the capital market. It reflects the terms on which creditors are willing to lend to those with control over the leveraged entity. One plausible channel through which the constraint operates is the wish by creditors to avoid being embroiled in a lengthy and costly bankruptcy settlement after the borrower has defaulted. When a bank breaches the maximum leverage ratio permitted by the market, the bank must take remedial action to reduce its leverage or face a run by its creditors.

We have seen that throughout the GFC, banks lost pure equity capital through credit losses and dividend payouts but did not replenish pure equity capital through the issuance of new common equity. Instead, the lion's share of new capital raised was in the form of hybrid claims such as preferred shares and subordinated debt. In particular, government-sponsored capital injections took the form of preferred equity. This was especially the case in the United States under TARP. The consequence was that pure equity capital continued to dwindle.

[12] Acharya and Viswanathan (2011) builds a model of funding of liquidity of financial institutions tied to such a risk-shifting problem.

It would be reasonable to posit that the continuing stringency in credit conditions reflects, at least in part, the lack of pure equity capital in the banking system. Even though the Basel capital requirements are reduced by the injection of hybrid claims, the pure equity capital requirement remains as stringent as ever. Without concerted efforts to relax the market-determined capital requirements that are pressing down on the banks, it would have been difficult to expect much headway in freeing up credit conditions toward greater willingness of the banks to extend credit during the period.

The distinction between Basel capital and pure equity capital can also be seen through the evolution of various bank leverage ratios. This examination underscores the earlier evidence that asset growth of banks during 2000–2006 was funded primarily through debt, especially through short-term debt, and not through buildup of common equity capital.

Figure 17 shows the leverage ratios for the 25 large financial firms in our sample—divided into commercial banks, investment banks, and GSEs—for the fiscal years 2000 through 2009. The numbers reported are averages within each division.

Figure 17: Leverage Ratios for 25 Large Financial Firms in the Data Set

Leverage ratios	Type of bank	2000–2006[a]	1Q07	2Q07	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09
Total debt/ shareholder equity	Commercial bank	5.65	5.52	7.13	6.14	6.79	4.94	6.71	5.24	7.70	4.71	6.00	4.30	5.37
	Investment bank	13.88	16.66	17.20	18.00	19.39	20.01	15.60	13.20	10.52	9.13	9.50	9.10	8.22
	GSE	30.31	21.22	23.40	22.66	21.62	27.58	30.03	−379.04	−37.42	−70.72	−704.68	−352.85	−142.56
Total assets/ common equity[b]	Commercial bank	17.77	19.83	24.61	19.49	22.51	24.00	25.06	20.66	27.93	25.29	23.84	19.29	19.55
	Investment bank	25.96	31.13	31.63	33.78	35.85	41.91	33.80	29.63	29.77	28.17	22.20	20.84	18.82
	GSE	39.23	29.20	33.08	33.30	41.85	68.23	85.28	40.40	18.19	15.37	14.56	12.85	10.83
Total assets/ shareholder equity	Commercial bank	16.86	16.75	20.48	17.87	22.11	20.26	21.85	18.38	23.54	17.06	19.18	15.30	17.18
	Investment bank	25.33	29.59	30.09	32.05	33.91	37.58	29.50	27.43	23.44	21.40	19.81	18.74	17.01
	GSE	32.32	23.10	25.49	24.68	23.57	29.85	32.40	−398.83	-38.51	−74.85	−761.04	-385.79	−156.84
Commercial paper/total assets	Commercial bank	0.72%	1.62%	0.82%	1.92%	3.00%	1.76%	0.83%	1.71%	2.46%	1.70%	0.67%	0.91%	1.10%
	Investment bank	0.57%	0.00%	0.18%	0.07%	2.88%	2.30%	0.00%	0.00%	3.22%	5.88%	3.19%	2.05%	3.88%
	GSE	0.12%	0.44%	0.37%	1.13%	1.11%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

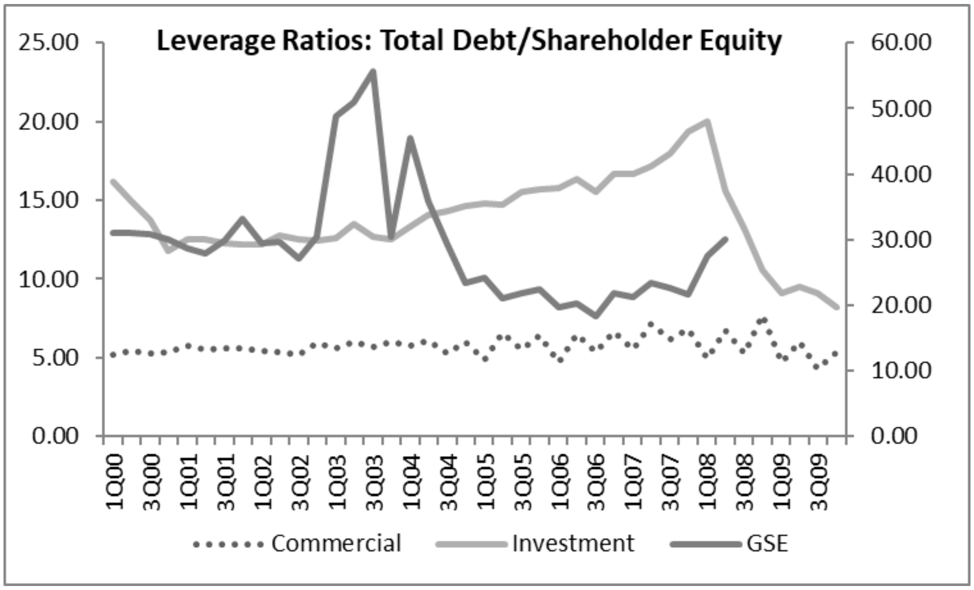
(a) The 2000–2006 numbers are averages of the quarterly ratios from 2000 through 2006.

(b) Common equity = shareholder equity – preferred equity as reported in balance sheets.

Sources: Balance sheets of all banks, from Bloomberg, US Securities and Exchange Commission filings, and annual reports.

Figures 18, 19, 20, and 21 are based on the time-series evolution of four of these ratios, which we focus on in our discussion.

Figure 18: Leverage Ratios—Total Debt/Shareholder Equity for Large US Financial Firms

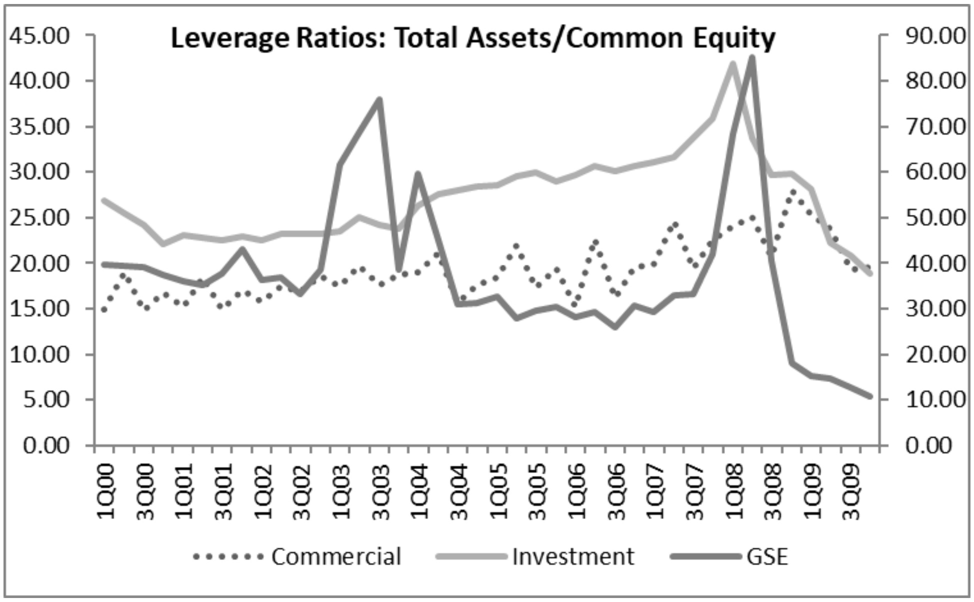


Note: Commercial and investment bank ratios are on the primary (left-side) axis, and GSE ratios are on the secondary (right-side) axis.

Total debt = short-term borrowings + long-term borrowings. It does not include deposits held by a bank.

Sources: Balance sheets of all banks from Bloomberg, US Securities and Exchange Commission filings, and annual reports.

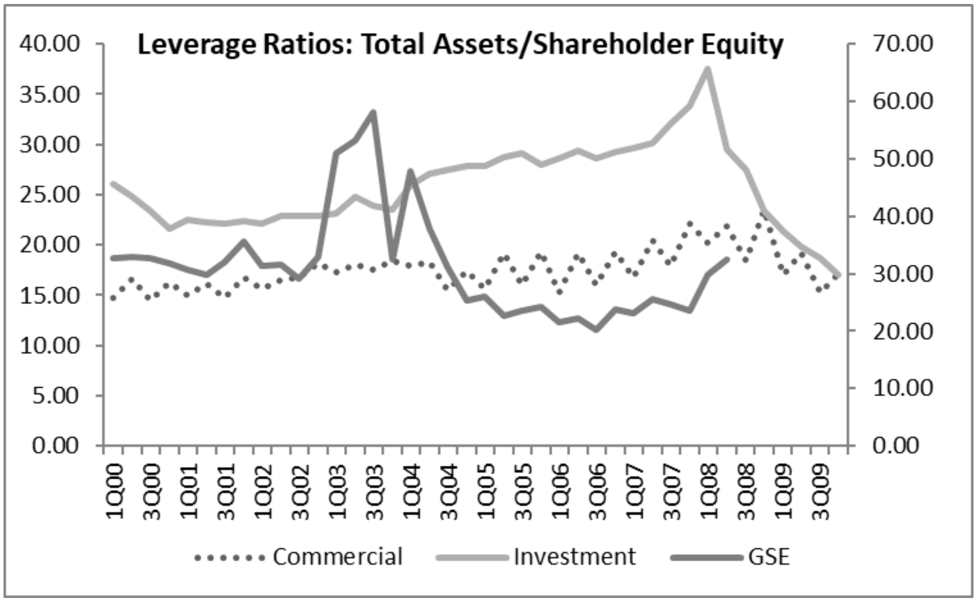
Figure 19: Leverage Ratios—Total Assets/Common Equity for Large US Financial Firms



Note: Commercial and investment bank ratios are on the primary (left side) axis, and GSE ratios on the secondary (right side) axis.

Sources: Balance sheets of all banks from Bloomberg, US Securities and Exchange Commission filings, and annual reports.

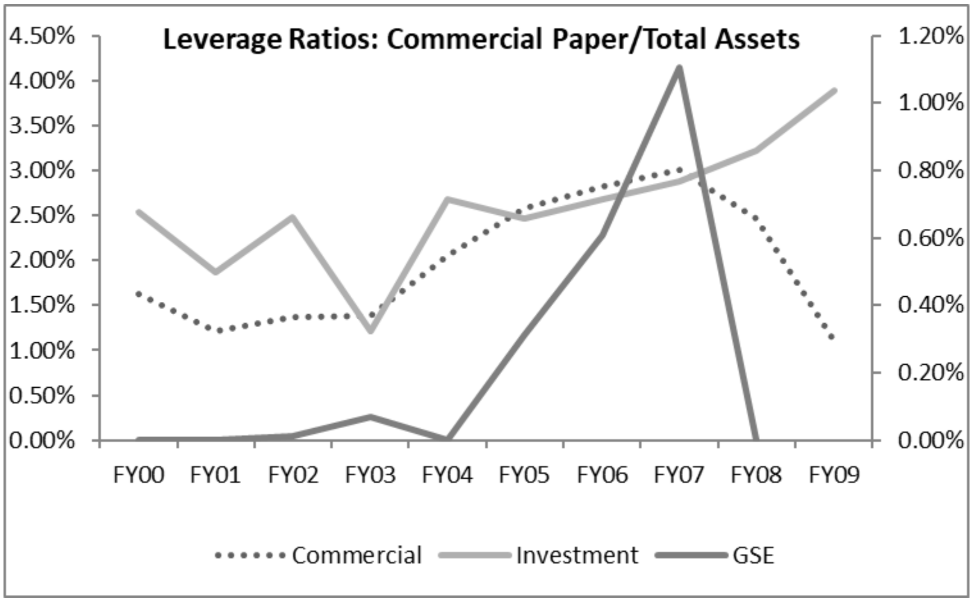
Figure 20: Leverage Ratios—Total Assets/Shareholder Equity for Large US Financial Firms



Note: Commercial and investment bank ratios are on the primary (left side) axis, and GSE ratios on the secondary (right side) axis.

Sources: Balance sheets of all banks from Bloomberg, US Securities and Exchange Commission filings, and annual reports.

Figure 21: Leverage Ratios—Commercial Paper/Total Assets for Large US Financial Firms



Note: Commercial and investment bank ratios are on the primary (left side) axis, and GSE ratios on the secondary (right side) axis.

Sources: Balance sheets of all banks, from Bloomberg, US Securities and Exchange Commission filings, and annual reports.

Figure 18 shows the corporate finance measure of leverage—the debt to shareholder equity ratio—and Figure 19 shows another measure—the assets to common equity ratio (common equity being shareholder equity minus preferred equity). In both cases, the pattern is similar and capital structure was getting increasingly levered from 2000 to 2007.

For commercial banks, this ratio increased from 14.68 to 22.11 from Q1 ’00 to Q3 ’08. For investment banks, the Q1 ’00 ratio was much higher at 26.13 and increased to 33.91 in Q4 ’07. For GSEs, this ratio decreased from 32.8 to 23.57 during the same period. To summarize, the levering up of commercial and investment banks is evident across the three ratios we have shown. For GSEs, the assets to common equity ratio captures this dynamic.

Figure 22 and the corresponding Figure 23 show the change in the assets-to-common-equity ratio during the crisis for the large US financial firms in our sample. This ratio increased from Q1 ’07 to the peak of the crisis in Q2 and Q3 ’08 for most firms, as is evident for Citigroup and Lehman before it went bankrupt. This increase in the assets-to-common-equity ratio was even more dramatic for the GSEs as they became distressed.

To sum, the analysis till now shows that the significant growth in banks’ assets during 2000 to 2007 was increasingly funded by debt. We now examine the type of debt funding this asset growth. To shed light on this question, we plot in Figure 21 the ratio of commercial paper to total assets for commercial banks, investment banks, and GSEs in our sample. While investment banks were always financed in a significant way through unsecured short-term commercial paper, what is striking is that commercial banks increased their reliance on commercial paper nine-fold from fiscal year 2000 to fiscal year 2007. In Q4 '07, commercial paper was 3% of assets for commercial banks relative to an average of 2.27% between 2000 to 2006. This is comparable to investment banks with a ratio of 2.88% in Q4 '07 relative to an average of 1.86% between 2000 to 2006.

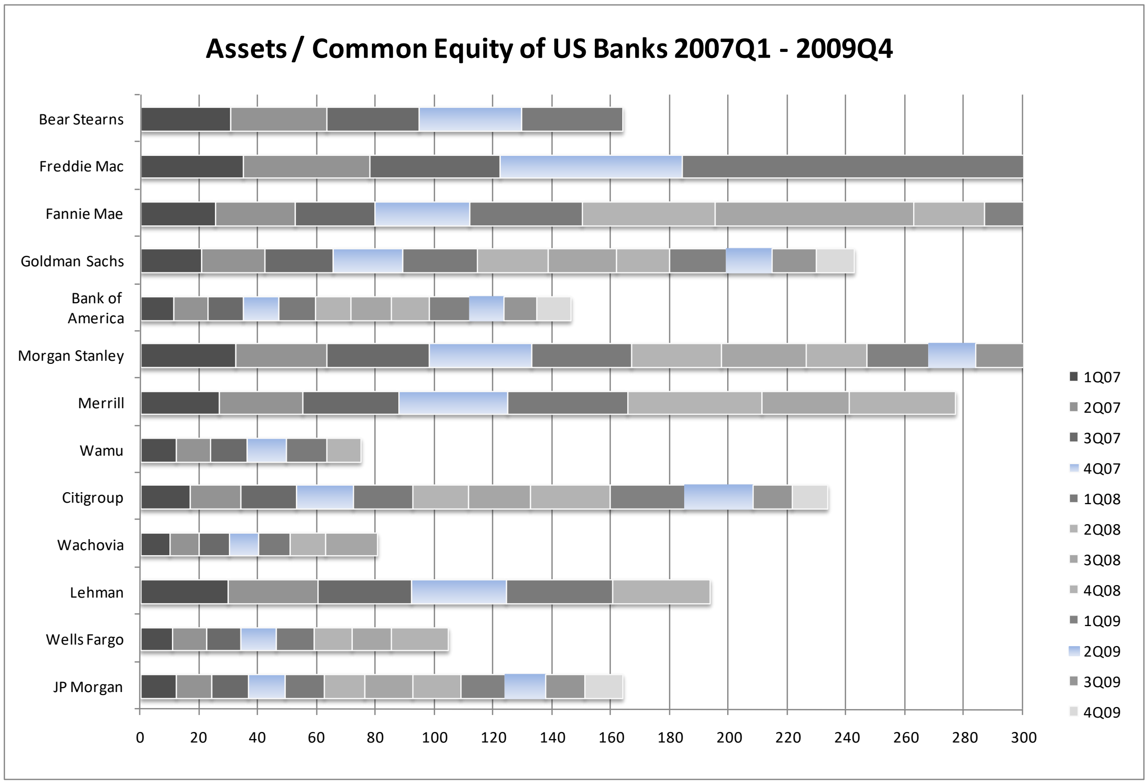
Further, while the growth in loans and assets was primarily of the long-term type—for mortgages to a large extent and corporate and private equity finance to some extent—the nature of nondeposit debt financing was of the short-term type. That is, bank capital structures were not only looking increasingly levered and funded through nondeposit-type debt, but they were also experiencing a rise in maturity mismatch (or duration gap between assets and liabilities) and were thus vulnerable to economy-wide shocks that generally tend to cripple the markets for short-term financing.

Figure 22: Assets/Common Equity Ratios for Large U.S Financial Firms, 2007–2009

Firm	1Q07	2Q07	3Q07	4Q07	1Q08	2Q08	3Q08	4Q08	1Q09	2Q09	3Q09	4Q09
JPMorgan	11.97	12.23	12.33	12.68	13.08	13.96	16.35	16.12	15.04	13.82	13.24	12.93
Wells Fargo	10.70	11.57	11.63	12.20	12.58	12.89	13.43	19.33	0.00	0.00	0.00	0.00
Lehman	29.73	30.24	31.94	32.30	35.99	33.16	0.00	0.00	0.00	0.00	0.00	0.00
Wachovia	9.69	9.94	10.27	10.05	10.74	11.89	17.71	0.00	0.00	0.00	0.00	0.00
Citigroup	16.69	17.47	18.58	19.26	20.21	19.27	20.78	27.32	25.43	23.15	13.24	12.00
Washington Mutual	12.06	11.71	12.51	13.06	13.92	11.64	0.00	0.00	0.00	0.00	0.00	0.00
Merrill	26.50	28.65	32.39	37.03	40.79	45.77	29.44	36.16	0.00	0.00	0.00	0.00
Morgan Stanley	32.07	31.24	34.70	34.65	33.90	30.88	28.48	20.80	20.87	16.17	18.05	17.86
Bank of America	11.38	11.55	11.69	12.05	12.49	12.39	13.38	13.05	13.96	11.47	11.32	11.45
Goldman Sachs	20.59	21.40	23.28	23.84	25.18	23.97	23.46	17.85	19.23	15.66	14.86	13.12
Fannie Mae	25.32	26.98	27.06	32.32	38.19	45.05	67.47	24.41	16.49	13.75	10.95	9.00
Freddie Mac	34.62	43.28	44.07	62.10	392.09	853.44	27.92	14.29	14.41	15.50	15.64	13.70
Bear Stearns	30.55	32.69	31.39	34.56	34.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Sources: Bloomberg, US Securities and Exchange Commission filings, and annual reports.

Figure 23: Assets to Common Equity Ratios of US banks, 2007–2009



Note: Figure does not depict ratios for Wells Fargo, Lehman, Wachovia, Washington Mutual, Merrill and Bear Stearns for all quarters due to unavailability of data.

Sources: Bloomberg, US Securities and Exchange Commission, annual reports, Datastream.

This short-term aspect of bank leverage is captured in Figure 24 and 26, and corresponding Figure 25. Figure 24 shows the worldwide quarterly outstanding amounts for commercial paper—usually of 90-day maturity and more than 75% of which tended to be issued by financial institutions. From a steady issuance of around $1.4 trillion–$1.5 trillion during 2000–2004, the amount rose sharply to a peak of $2.15 trillion during Q2 '07. Following the money market freeze of August 9, 2007, this figure fell sharply from its peak to about $1.62 trillion in Q3 '08. It then picked up somewhat in Q4 '08 due to guarantees, for example, by the Federal Reserve. In 2009, commercial paper issuance declined further, and in Q4 '09, this figure was about $1.15 trillion.

Viewed from any dimension—overall leverage, deposit versus nondeposit leverage, and maturity of leverage—banks were pursuing a risk-shifting strategy, and importantly, not just through their choice of assets but also through their capital structures. There is one important lesson for bank regulation in these findings. We highlight that standard corporate finance measures of capital, dividend distribution, and leverage were individually and jointly implying that bank behavior reflected a serious conflict of interest between shareholders and creditors. At the same time, regulatory measures of capital adequacy—for example, the ratio of capital to risk-adjusted assets—hardly moved during the period[13]. Why was this so? While some of this had to do with the large holdings of AAA-rated tranches of mortgage-backed securities on bank balance sheets, which attracted little capital charge and thus kept the level of risk-adjusted assets (the denominator) to a low figure, the measurement of capital (the numerator) was also problematic.

Figure 24: Commercial Paper Outstanding Worldwide—Quarterly Data for Commercial Paper (in $ bn)

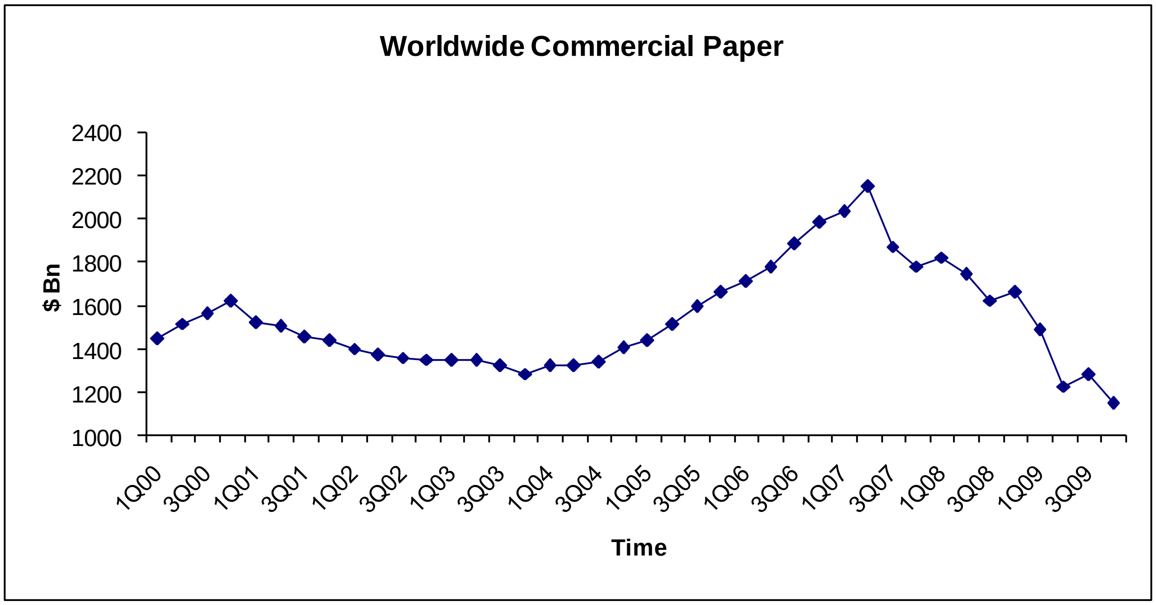
Quarter	CP	Quarter	CP	Quarter	CP	Quarter	CP
1Q00	1,449.1	**3Q02**	1,360.8	**1Q05**	1,436.9	**3Q07**	1,871.8
2Q00	1,517.2	**4Q02**	1,352.3	**2Q05**	1,514.7	**4Q07**	1,780.6
3Q00	1,560.2	**1Q03**	1,349.9	**3Q05**	1,597.2	**1Q08**	1,821.5
4Q00	1,619.3	**2Q03**	1,349.8	**4Q05**	1,662.0	**2Q08**	1,741.1
1Q01	1,523.0	**3Q03**	1,321.4	**1Q06**	1,709.9	**3Q08**	1,624.3
2Q01	1,504.4	**4Q03**	1,284.2	**2Q06**	1,776.4	**4Q08**	1,658.8
3Q01	1,457.0	**1Q04**	1,323.5	**3Q06**	1,886.0	**1Q09**	1,488.8
4Q01	1,437.4	**2Q04**	1,323.0	**4Q06**	1,982.9	**2Q09**	1,229.1
1Q02	1,400.2	**3Q04**	1,341.2	**1Q07**	2,034.7	**3Q09**	1,279.5
2Q02	1,372.6	**4Q04**	1,403.8	**2Q07**	2,149.7	**4Q09**	1,147.7

Note: Commercial paper outstanding is seasonally adjusted.

Source: Bloomberg.

[13] See the detailed analysis in Box 1.3 of International Monetary Fund 2008, a report that looks at the 10 largest publicly listed banks in the U.S. and Europe.

Figure 25: Quarterly Data for Commercial Paper Worldwide ($ bn)



Source: Bloomberg.

Figure 26: Commercial Paper Issued by 25 Large Financial Firms

(in $ bn)	Firm	Type of firm	FY00	FY01	FY02	FY03	FY04	FY05	FY06	FY07	FY08	FY09
US	JPMorgan	Commercial bank	24.9	18.5	16.6	14.0	12.6	13.9	18.8	49.6	37.8	41.8
US	Wells Fargo	Commercial bank	15.8	14.0	11.1	6.7	6.2	4.0	1.1	30.4	45.9	13.0
US	Lehman Brothers	Investment bank	4.2	1.9	1.6	1.6	1.7	1.8	1.7	3.1	0.0	0.0
US	Wachovia Corp.	Commercial bank	2.9	2.9	3.1	7.2	12.0	3.9	4.7	6.7	0.0	0.0
US	Citigroup	Commercial bank	18.7	13.9	18.3	17.6	25.6	34.2	43.7	37.3	28.7	10.2
US	Washington Mutual	Commercial bank	1.0	0.4	0.7	1.1	4.0	7.1	4.8	2.0	0.0	0.0
US	Merrill Lynch	Investment bank	13.0	1.9	3.4	3.4	4.0	3.9	6.4	12.9	20.1	0.0
US	Morgan Stanley	Investment bank	27.8	32.8	50.8	28.4	28.5	23.2	22.4	22.6	6.7	0.8
US	Bank of America	Commercial bank	7.0	1.6	25.2	42.5	78.6	116.3	141.3	191.1	158.1	0.0
US	Goldman Sachs	Investment bank	10.7	8.4	9.5	4.8	4.4	5.2	1.5	4.3	1.1	62.5
US	Fannie Mae	GSE	0.0	0.0	0.2	1.3	0.0	5.1	10.0	0.0	0.0	0.0
US	Freddie Mac	GSE	0.0	0.0	0.0	0.0	0.0	0.0	0.0	18.5	0.0	0.0
US	Bear Stearns	Investment bank	0.0	0.0	0.0	0.0	0.0	0.0	20.7	3.9	0.0	0.0
UK	Royal Bank of Scotland	Commercial bank	1.0	0.4	11.2	6.3	16.1	25.1	24.8	155.9	71.0	41.4
UK	HSBC	Commercial bank	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
UK	Barclays Plc	Commercial bank	0.0	4.8	8.4	7.9	40.1	50.4	51.9	46.5	40.4	31.2
UK	HBOS	Commercial bank	2.0	11.2	15.0	23.0	0.0	0.0	33.9	33.5	129.9	0.0
UK	Lloyds TSB	Commercial bank	0.0	0.0	0.0	0.0	15.4	18.6	25.6	34.5	42.2	56.6
Europe	IKB	Commercial bank	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Europe	UBS	Investment bank	0.0	0.0	0.0	0.0	69.7	77.8	98.0	133.6	104.1	49.9
Europe	Credit Suisse	Commercial bank	0.0	0.0	0.0	0.0	0.0	7.9	12.3	13.0	4.5	4.8
Europe	Deutsche Bank	Commercial bank	0.0	0.0	0.0	16.5	13.5	15.9	43.0	42.7	36.5	31.5
Europe	Fortis Bank	Commercial bank	0.0	0.0	0.0	0.0	60.4	78.5	100.4	109.2	0.0	0.0
Europe	BNP Paribas	Commercial bank	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Europe	ABN AMRO	Commercial bank	26.0	20.3	25.1	20.3	20.9	61.5	74.4	63.3	41.9	30.3
TOTAL	—	—	**154.8**	**132.8**	**200.1**	**202.5**	**413.8**	**553.9**	**741.4**	**1014.6**	**768.9**	**374.0**

Note: Commercial paper information is as reported in the cash flow statements of financial statements from Bloomberg or US Securities and Exchange Commission filings. This data is not available on Bloomberg or US Securities and Exchange Commission filings for the following institutions and years: HSBC and BNP for all years; HBOS in 2004–2005 and 2009; Barclays in 2000; and Lloyds TSB, UBS, Credit Suisse in 2000–2004, Deutsche Bank, Fortis Bank in 2000–2003 and 2008–2009, and all years for IKB, BNP Paribas.

Sources: Bloomberg, US Securities and Exchange Commission filings, and annual reports.

IV. Why Don't Banks Cut Dividends during Crisis Periods?

Consistent with the summary statistics presented in Section II, anecdotal evidence also highlights the reluctance of banks to cut dividends or even reduce the amount of dividends during the GFC.[14] Lehman Brothers Holdings announced a 13% increase in its dividend and a $100 million share repurchase in January 2008. Citigroup cut its dividend close to zero only

[14] See Figures 12, 13, 14, and 15 for bank-by-bank histories of dividend distributions. Also, see related press articles: Eckblad and Barris 2009, Lawder 2009, Miller 2009, and Reuters 2009.

in November 2009. JPMorgan and Wells Fargo, while recipients of TARP capital in fall 2008, cut dividends as late as February and March 2009, respectively. In addition, even as the Federal Reserve was urging banks receiving bailout funds to cut dividends, Goldman Sachs and Morgan Stanley did not cut dividends throughout the crisis period.

This is to be compared to the fact that 61 components of the Standard & Poor's 500 stock index cut their dividends during 2008. Most corporate debt has covenants that prevent firms from paying out dividends when negative earnings are reported. This constraint prevents firms from transferring funds to equity holders at the expense of debt holders.

In contrast, banks continued to pay out dividends even during the crisis. This can be attributed to the short-term nature of their funding and the implicit and explicit guarantees provided by the government. Banks are typically funded by short-term debt. As a result, if they were to announce a dividend cut, rollover debt could "run" as it did on investment banks. The fear of "runs" leads banks to continue paying dividends even when it would be prudent for them in the long run to cut dividends.

Further, banks benefit from the explicit and implicit guarantees provided by the government. The explicit government guarantees provided on deposits for commercial banks ensure that the banks are protected even in the event of a failure.[15] Similarly, many financial institutions may have the implicit government guarantee for firms that are considered too big to fail. Thus, banks are unlikely to cut dividends, figuring that in the event that they do fail, they would most likely be bailed out.[16]

The contrast between stressed depository institutions (such as Wachovia and Washington Mutual) and investment banks (such as Lehman and Merrill Lynch) is informative. While depositories were subject to a "prompt corrective action" resolution regime, such orderly wind-down plans were absent for investment banks. Hence, the implicit too-big-to-fail guarantee for investment banks was virtually free of any endgame restrictions, allowing them to pay dividends even as they were failing.

The contrasting behavior of banks versus nonfinancial firms provides important lessons for reform of governing bank regulation. Regulators have realized that banks need to be explicitly prevented from paying out dividends in times of distress to avoid such transfers in violation of the priority of debt over equity. As Lawrence Summers, then director-designate, National Economic Council, noted in his January 12, 2009, letter to the Senate and House of Representatives (Bloomberg 2009):

---

[15] Countries in the European Union have different amounts of deposit insurance guaranteed, with the minimum amount of EUR 50,000 in 2008. The UK guaranteed GBP 85,000 for single accounts, whereas the U.S. guaranteed $250,000 per person.

[16] Such too-big-to-fail guarantees can also apply to large firms. General Motors, Chrysler, and Ford—the Big Three automakers in the US—were bailed out by Congress post the 2008 financial crisis, as their bankruptcies could have led to a loss of 1 million jobs. See https://www.thebalance.com/auto-industry-bailout-gm-ford-chrysler-3305670.

> Those receiving exceptional assistance will be subject to tough but sensible conditions that limit executive compensation until taxpayer money is paid back, ban dividend payments beyond de minimis amounts, and put limits on stock buybacks and the acquisition of already financed strong companies.

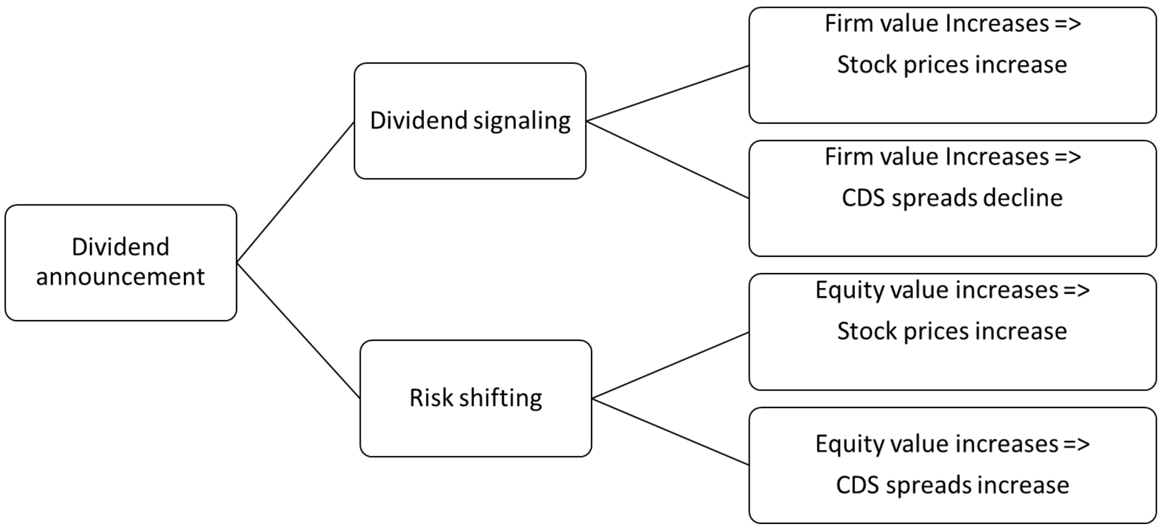
Although this is a step in the right direction, more needs to be done. In later sections, we further draw on the lessons learnt from private contracting and provide recommendations for the design of prompt corrective measures for governing bank regulation.

Empirical Evidence for the Signaling versus Risk-Shifting Story

Now, we try to distinguish between two possible explanations as to why banks increased or did not cut dividends. In a dividend-signaling story, issuance of a dividend increase should be perceived as good news for the firm (both debt and equity). Hence, the share price and creditor reaction should be positive around the announcement of an increase in dividends. Typical dividend signaling models are, however, without leverage or risk-shifting considerations. In a risk-shifting story, issuance of a dividend increase should be perceived as good news for only equity but not debt. If the signal being conveyed is about the value of the firm rather than an attempt to just raise the value of the equity (as in the risk-shifting story), then the news of the announcement should be good news for creditors of the firms as well. In contrast, if there is no good news about the firm (or potentially the news is bad for the firm), but simply good for equity, then the announcement should benefit shareholders but hurt the creditors as it is a transfer of value across firm claimants rather than overall increase in value of the firm.

Thus, examining debt price reactions around dividend change announcements can shed light on the question of whether the dividend changes are purely a signaling attempt or (also) an attempt to shift risks toward creditors. Figure 27 summarizes the empirical tests analyzing the impact of a dividend announcement on stock prices, and credit default swap (CDS) spreads can be used to empirically distinguish between the risk-shifting hypothesis versus the dividend-signaling hypothesis of why banks continued to pay dividends during the GFC. Note, one caveat in interpreting stock price and CDS spread reaction is that government guarantees and the presence of the Federal Deposit Insurance Corporation (FDIC) interventions can distort creditor responses.

Figure 27: Distinguishing between the Dividend Signaling versus Risk-Shifting Hypotheses



Source: Created by the authors.

We exploit the stock market and creditor response for investment bank versus commercial bank differences to potentially get around this problem. Specifically, we look at the abnormal returns after a dividend increase and no reduction versus dividend cuts, separately for investment and commercial banks. Then, we repeat the analysis for CDS spreads. This will help us determine the value to equity versus creditors and thus possibly differentiate a risk-shifting story from a pure signaling story.

In Figure 28, we see that relative to the market, stock prices increased by 0.5% after the announcement of a dividend increase (or no change in dividends) for commercial banks and by 0.6% for investment banks. Similarly, a dividend cut was accompanied by a stock price decline of 0.3% for commercial banks and a striking 3.1% decline for investment banks. This finding, an increase in stock prices when dividends were increased or maintained and a decline in stock prices when dividends were cut, is consistent with both a signaling story as well as a risk-shifting story.

We now turn to how the creditors responded to distinguish between the two alternative hypotheses. Figure 28 shows that CDS spreads (relative to the average market CDS) declined by 1.19 basis points (bps) for commercial banks when they increased (or maintained) their dividends, indicating that the creditors of commercial banks perhaps believed that firm value increased. This is consistent with a signaling story for commercial banks. In contrast, a dividend increase for investment banks was accompanied by an increase in CDS spread of 5.85 bps, indicating that equity value increased at the expense of creditors. This is consistent with a risk-shifting story for investment banks. The findings on creditor response to dividend cuts were distorted by the presence of government guarantees and FDIC intervention for commercial banks. We see that a dividend cut for investment banks was accompanied by a lower increase in CDS spreads of 3 bps. In contrast, a dividend cut for commercial banks resulted in a decrease in CDS, which is not completely consistent with a signaling story.

Creditors may have priced in the fact that the presence of government guarantees and FDIC interventions for commercial banks implied that the banks were protected even in the event of a failure.

Figure 28: CDS Spreads and Stock Prices

Stock Prices

Bank type	Dividend raises/neutral	Dividend cuts	Observations
Commercial	0.005	–0.003	74
Investment	0.006	–0.031	29

CDS Spreads

Bank type	Dividend raises/neutral	Dividend cuts	Observations
Commercial	–1.186	–4.104	39
Investment	5.854	2.962	22

Note: The table above reports the average stock return (top) and CDS spread (bottom) relative to the market return on the day of the dividend announcement. This is measured separately for dividend raises/neutral and dividend cuts separately for investment and commercial banks. In the top panel, the abnormal returns are the fraction, and in the bottom panel, the spreads are in basis points. We look at the return on the day of the dividend announcement for the period from September 2007 to December 2009. Stock price data for all banks is from Bloomberg. CDS spread data is from Datastream. We separately analyze for dividend increases and dividend decreases.

Sources: Bloomberg, Datastream.

To summarize, the differential response of creditors of commercial banks relative to investment banks provides evidence of a risk-shifting story for investment banks and a signaling story for commercial banks. We see that a dividend increase for commercial banks resulted in a decrease in CDS spreads. In contrast, for investment banks, a dividend increase resulted in an increase in CDS spreads. When we look at stock prices, an increase in dividends by banks was associated with an increase in stock price for both commercial and investment banks. In Section II, indeed, we saw that the dividend-increase problem in the face of adverse funding conditions was predominantly a feature for investment banks, and commercial banks often cut or maintained dividends (which could potentially be due to the fear of FDIC interventions).

V. Implications for Reform of Financial Regulation

The distinction between Basel capital and pure equity capital emphasized in this paper has important implications for the reform of financial regulation and the resolution of problematic banks that can lead to a speedy recovery in lending.

To some extent, some inertia is inevitable in the valuation of bank assets, even in a world where the rigorous application of mark-to-market valuation rules are the preferred norm.

Even under the original version of accounting standards such as the US accounting standard 157 of the Financial Accounting Standards Board (FASB), or the International Accounting Standards Board (IASB) rule IAS 39, full and immediate marking to market of assets is infeasible due to the lack of transparent markets. There is also the larger issue of whether full marking to market is even desirable from a financial stability viewpoint. Here, we will not address this particular debate. However, even for a fervent supporter of full marking to market as an ideal, the practical limitation of marking to market of bank assets means that inertia is an inevitable feature of bank balance sheet accounting.

In a world where bank balance sheets lag market conditions, or where the accounting values do not anticipate further credit losses from foreseeable weakening of macroeconomic activity, an early suspension of dividends and capital preservation would seem to be one of the first steps that a regulator must take to forestall greater problems with capital erosion in the future. The FDIC could replicate a "covenant" style private contract, which restricts banks from paying out dividends when certain thresholds are reached. There should be an explicit role in the covenant thresholds for simple leverage measures such as assets-to-common-equity ratios, loans to deposits, and short-term debt to assets. Additionally, market measures such as equity retention implied by repo haircuts may provide more timely information and prevent further equity erosion by forcing banks to stop paying out dividends in times of distress.

From the point of view of overall financial system stability and the externalities imposed by one constituency on the system as a whole, an early suspension of dividends can be justified to prevent incumbent controlling shareholders from imposing negative spillover effects from weakened banks on the rest of the system. Although such interference in the management of the firm runs counter to the autonomy of the controlling shareholders in determining the financial decisions of the firm, it should be borne in mind that banking has always offered exceptions to this rule (Dow 1996).

The fact that banks have been regulated reflects their special status. They exert externalities on the rest of the financial system so that the affairs of the bank affect a very broad constituency that goes beyond the traditional domain of the owners and creditors of the firm. They affect the broader economy and are supported by both explicit and implicit public funding support in case of difficulties. Our proposal for an early suspension of dividends is merely redrawing the line between the private and public domains of actions. Such suspensions could be triggered once signals of market stress are patently visible to regulators, such as stress in interbank markets, difficulty in rolling over wholesale finance, accelerated drawdowns of bank credit lines, a sharp rise in market volatility, or significant declines in bank market valuations. Regulatory action also requires that banks conduct timely stress tests to determine whether banks can absorb losses during crises.

The onset of the COVID-19 pandemic in 2020 saw regulators in the US, the UK, and Europe impose restrictions on dividends in some form. For example, in March 2020, the European Central Bank (ECB) asked that banks halt cash dividends and share buybacks, citing the need to preserve banks' capacity to absorb these losses.[17] In the UK too, the Bank of England asked its seven biggest lenders to suspend dividends until the end of 2020.[18] The US Federal Reserve imposed similar restrictions citing the need to conserve capital during the lockdown-induced downturn.[19] Thus, an early imposition of regulatory sanctions against the paying of dividends (for instance, as part of an increasing "ladder of sanctions" that use market or common equity based notions of bank leverage) may have an important place in the agenda for reform of the regulatory system. The proposals in the Geneva Report (Brunnermeier et al., 2009) argue for such a ladder of sanctions. Acharya, Mehran, and Thakor (2010) suggest creating a capital account by diverting dividends during good times which are then transferred to a regulator when the bank goes bankrupt.

VI. Conclusion

In this paper, we have delved deeper into the evolution of bank capital during the Global Financial Crisis. The crisis, which initially erupted in 2007 in the subprime mortgage sector in the United States, led to a decline in real economic activity, leading to further credit losses in other mainstream credit categories such as prime mortgages, commercial real estate, corporate debt, and other household debt such as credit card loans and auto loans.

Even as banks and financial intermediaries suffered large credit losses as the financial crisis gathered pace, the headline numbers obscured important shifts in the composition of bank capital, and hence on the constraints banks faced in their daily operations. We have shown that the bulk of the new capital raised both from private investors and from government-funded capital injections was in the form of debt-like hybrid claims such as preferred equity and subordinated debt, and not in the form of common equity. Furthermore, banks continued to pay large sums in the form of dividends, which further eroded the common equity base.

As a result, there was a relentless increase in the leverage of the banking sector, when leverage is measured with common equity as the denominator. Notably, total assets to common equity for commercial banks increased from an average of 17.77 between 2000 and 2006 to 22.27 in 2007–2009. Additionally, the ratio for investment banks increased from 25.96 in 2000–2006 to 29.79 in 2007–2009. We have argued that common equity, or what we have termed pure equity capital, is the more appropriate notion of bank capital when we want to capture the idea of market-based capital requirements that creditors would like to impose on borrowers. The alternative notion of bank capital, termed Basel capital, which includes subordinated debt and hybrid claims (as a buffer against loss for depositors) is less

---

[17] For the ECB announcement, see European Central Bank 2020.

[18] See https://www.thenationalnews.com/business/banking/bank-of-england-relaxes-restrictions-on-lenders-paying-dividends-1.1126403.

[19] See https://www.reuters.com/article/usa-fed-banks-idUSL1N2LN3FW for the Federal Reserve's announcement.

appropriate, even though this latter notion of capital is what is enshrined in the current banking regulations.

We argue that continuing dividend payments during the crisis represented a transfer from creditors (and taxpayers) to equity holders of banks in violation of the priority of debt over equity. We further argue that the increased riskiness of the remaining assets of the banks represented a type of risk-shifting that benefited equity holders at the expense of creditors (and taxpayers).

In general, the events of the financial crisis of 2007–2009 have posed several challenging questions on the proper notion of bank capital that should inform bank regulation. We offer our paper as a small step in this important debate. With the onset of the COVID-19 pandemic, regulators in the US, the UK, and Europe have imposed dividend restrictions on banks in their countries. An interesting avenue for future research could explore the efficacy and timeliness of such dividend suspensions.

VII. References

Acharya, Viral V., Hamid Mehran, and Anjan V. Thakor. 2010. *Caught between Scylla and Charybdis? Regulating Bank Leverage When There Is Rent Seeking and Risk Shifting*. Federal Reserve Bank of New York Staff Report No. 469.

Acharya, Viral V., Philipp Schnabl, and Gustavo Suarez. 2009. *Securitization without Risk Transfer*. Working Paper. New York University Stern School of Business.

Acharya, Viral V., and S. Viswanathan. 2011. "Leverage, Moral Hazard, and Liquidity." *Journal of Finance* 66, no. 1: 99–138.
https://doi:10.1111/j.1540-6261.2010.01627

Adrian, Tobias, and Hyun Song Shin. 2010. "Liquidity and Leverage." *Journal of Financial Intermediation* 19, no. 3: 418–37.
https://doi.org/10.1016/j.jfi.2008.12.002

Bae, Kee-Hong, Jun-Koo Kang, and Jin Wang. 2011. "Employee Treatment and Firm Leverage: A Test of the Stakeholder Theory of Capital Structure." *Journal of Financial Economics* 100, no. 1: 130–53.
https://doi.org/10.1016/j.jfineco.2010.10.019

Bloomberg. 2009. "Obama to act tough on Banks in rescue package." *Business Standard*, January 29, 2013.
https://www.business-standard.com/article/economy-policy/obama-to-act-tough-on-banks-in-rescue-package-109012601015_1.html

Brunnermeier, Markus, Andrew Crockett, Charles Goodhart, Avinash Persaud, and Hyun Song Shin. 2009. *The Fundamental Principles of Financial Regulation*. Geneva: International Center for Monetary and Banking Studies Centre for Economic Policy Research.

Diamond, Douglas W., and Raghuram G. Rajan. 2011. "Fear of Fire Sales and the Credit Freeze." *Quarterly Journal of Economics* 126, no. 2: 557–91. May.
https://doi.org/10.1093/qje/qjr012

Dudley, William C. 2009. "Financial Market Turmoil: The Federal Reserve and the Challenges Ahead." *Council on Foreign Relations Corporation Conference.* Speech presented at the Council on Foreign Relations Corporate Conference on March 6, 2009, at the New York City

Eckblad, Marshall, and Mike Barris. 2009. "Wells Fargo Cuts Its Dividend 85%." Wall Street Journal, March 7, 2009.
https://www.wsj.com/articles/SB123634454649952469

European Central Bank. 2020."ECB asks banks to refrain from or limit dividends until September 2021." Press release, December 15, 2020.
https://www.bankingsupervision.europa.eu/press/pr/date/2020/html/ssm.pr201215~4742ea7c8a.en.html

Faleye, Olubunmi, Vikas Mehrotra, and Randall Morck. 2006. "When Labor Has a Voice in Corporate Governance." *Journal of Financial and Quantitative Analysis* 41, no. 3: 489–510. doi:10.1017/S0022109000002519

International Monetary Fund. 2008. "Containing Systemic Risk and Restoring Financial Soundness." *International Monetary Fund Global Financial Stability Report*. April. https://www.imf.org/en/Publications/GFSR/Issues/2016/12/31/Containing-Systemic-Risks-and-Restoring-Financial-Soundness

Ivashina, Victoria, and David S. Scharfstein. 2010. "Bank Lending during the Financial Crisis of 2008." *Journal of Financial Economics* 97, no. 3: 319–38. https://doi.org/10.1016/j.jfineco.2009.12.001

Jensen, Michael C., and William H. Meckling. 1976. "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure." *Journal of Financial Economics* 3, no. 4: 305–60.

Lawder, David. 2009. "Fed to banks: don't use bailout funds for dividends." Reuters, February 24, 2009. https://www.reuters.com/article/us-usa-fed-dividends-idUSTRE51N5O720090224

Miller, Jay. 2009. "Citigroup Cuts Dividend." Wall Street Journal, January 20, 2009. https://www.wsj.com/articles/SB123248865074399621

Myers, Stewart C. 1977. "Determinants of Corporate Borrowing." *Journal of Financial Economics* 5, no. 2: 147–75.

Myers, Stewart C., and Nicholas S. Majluf. 1984. "Corporate Financing and Investment Decisions When Firms Have Information Investors Do Not Have." *Journal of Financial Economics* 13, no. 2: 187–221.

Reuters. 2009. "JPMorgan Cuts Dividend 87 Percent." *New York Times*, February 23, 2009. https://www.nytimes.com/2009/02/24/business/24morgan.html#:~:text=JPMorgan%20said%20its%20decision%20to,Troubled%20Asset%20Relief%20Program%20faster

Rosengren, Eric S. 2010. "Dividend Policy and Capital Retention: A Systemic 'First Response'." Speech presented at "Rethinking Central Banking" Conference, Washington, DC, October 10, 2010.

Saidenberg, Marc R., and Til Schuermann. 2003. "*The New Basel Capital Accord and Questions for Research 1.*" (2003). http://d1c25a6gwz7q5e.cloudfront.net/papers/1099.pdf

Scharfstein, David S., and Jeremy C. Stein. 2008. "This Bailout Doesn't Pay Dividends." *New York Times*, October 20, 2008.

Summers, Lawrence. 2009. Letter to Senate and House of Representatives on January 12, 2009.

Tucker, Paul. 2008. "Monetary Policy and the Financial System." Speech presented at Institutional Money Market Funds Association Annual Dinner, London, UK, April 2, 2008. https://www.bis.org/review/r080403e.pdf

Wessel, David. 2008. "Brainstorming about Bailouts." *Wall Street Journal*, March 13, 2008. http://online.wsj.com/article/SB120536045253831681.html

VIII. Appendixes

Appendix A: Variable Definitions

Main variables	Definitions	Source(s)
Credit losses & write-downs	Write-downs include those that directly reduce income, as well as value reductions that decrease only equity and are excluded by the banks from their earnings figures. The values are net of financial hedges the companies use to mitigate losses.	Bloomberg, WDCI function
Capital-raised (WDCI)	Capital infused by all banks, brokerages, insurance companies, and GSEs by different means.	Bloomberg, WDCI function
Capital raised	Includes net capital raised by long-term borrowings, net common equity issuance, and net preferred shares issued.	Bloomberg, SEC, annual reports, Datastream
Net capital	Includes net capital raised by long-term borrowings, net common equity issuance, and net preferred shares issued, less dividends.	Bloomberg, SEC, annual reports, Datastream
Dividend	Dividends paid in cash by banks.	Bloomberg, SEC, annual reports, Datastream, Compustat
Common equity	Common equity was calculated by subtracting preferred equity from total shareholder equity. Both preferred and shareholder equity numbers were taken from the banks' balance sheets.	Bloomberg, SEC, annual reports, Datastream
Profit & loss	Profit & loss of the bank as reported on the income statement.	Bloomberg, SEC, annual reports, Datastream
Assets	Total assets of the bank as reported on the balance sheet.	Bloomberg, SEC, annual reports, Datastream
Liabilities	Total liabilities of the bank as reported on the balance sheet.	Bloomberg, SEC, annual reports, Datastream
Total debt (leverage ratios)	Short-term borrowings + long-term borrowings as reported on the balance sheet. This does not include deposits held by banks.	Bloomberg, SEC, annual reports, Datastream
Loans	Loans + mortgages as reported on the balance sheet.	Bloomberg, SEC, annual reports, Datastream

Appendix B: Frequency of Data

Banks	Frequency	Years
US banks	Quarterly and annual information	2000-2009
European banks	Quarterly/Semiannual and annual information	2000-2009
UK banks	Semiannual and annual information	2000-2009